Process documentation: before and with the system

For each of the 25 modules, how the clinic did the work before (AS-IS, from the clinic interview only) and how the Multi-Branch Dental Clinic Operation System does it now (TO-BE, from the code and its tests).

Version of 2026-10-19 · branch feat/auth-messaging-paymongo · sources: docs/defense/INTERVIEW_EVIDENCE.md, docs/defense/MODULES_FINAL.md, docs/defense/AUTOMATION_EVIDENCE.md

Scope and how to read this document

- **AS-IS** steps come only from the interview: each shows its source (Q = first interview, PI = follow-up of 2026-10-11, MC = team confirmation of 2026-10-12). A step the interview did not describe is a **dashed amber box "to confirm with clinic"**; its question is in SECOND_INTERVIEW_QUESTIONS.md. Nothing about the old process is invented.
- **TO-BE** steps are what the system does today. Planned work is dashed and tagged "Planned"; browser-only prototypes are tagged "Mock only".
- No time savings are claimed: the interview gave no task durations, so every estimate says "not measured".
- Statuses: Implemented 11, Partial 9, Mock only 2, Planned 3.

Step from the interview (citation shown) · To confirm with the clinic · The system does it · A person does it in the system · Mock only · Planned (not built)

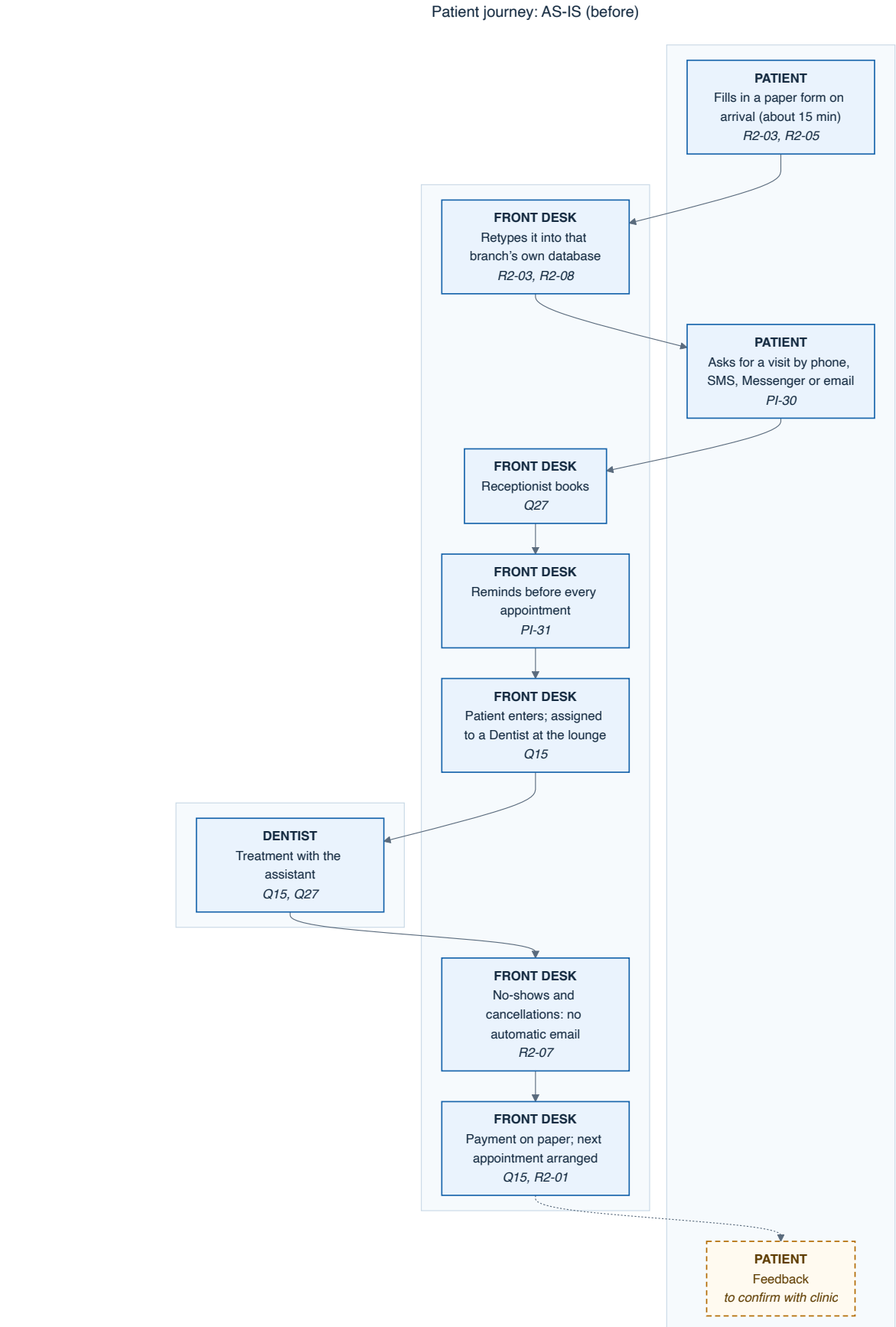
Module summary

#	Module	Status	Automation type
M1	User Management	Partial	Event-driven (code and invitation emails), rule-based (role checks)
M2	Branch Management	Partial	Rule-based
M3	Work Schedules	Implemented	Rule-based
M4	Patient Records	Partial	Rule-based (audit, name snapshot)
M5	Treatment Management	Implemented	Rule-based (lock), event-driven (to billing)
M6	Appointment Management	Implemented	Rule-based (open times, no overlaps), event-driven (emails), scheduled (reminders, waitlist)
M7	AI Chatbot (Clinic Assistant)	Implemented	Rule-based (knowledge matching)
M8	Patient Visits	Partial	Rule-based, scheduled (no-show flag; a person decides)
M9	Patient Queue	Partial	Rule-based (numbering)
M10	Dental Charts	Planned	Planned
M11	Billing Management	Implemented	Event-driven (draft, emails), rule-based (immutable charges), scheduled (overdue reminders)
M12	Treatment Plans	Planned	Planned
M13	Services and Pricing	Partial	Rule-based (effective price)
M14	Document Management	Partial	Rule-based
M15	Leave Management	Implemented	Rule-based
M16	Patient Inquiries	Partial	Rule-based
M17	Patient Messaging	Implemented	Event-driven (real-time, alert email)
M18	Notifications	Implemented	Event-driven, scheduled (reminders, retries)
M19	Prescriptions	Mock only	Planned
M20	Patient Follow-Up	Mock only	Scheduled recall email (off by default); follow-up tasks planned
M21	Reports and Analytics	Partial	Rule-based
M22	Audit Logs	Implemented	Event-driven
M23	Payment Management	Implemented	Integration (PayMongo), event-driven (receipt)
M24	Lab Cases	Planned	Planned
M25	Patient Handovers	Implemented	Rule-based

End-to-end patient journey: before (AS-IS)

Step from the interview (citation shown) · To confirm with the clinic · The system does it · A person does it in the system · Mock only · Planned (not built)

AS-IS (from the interview)

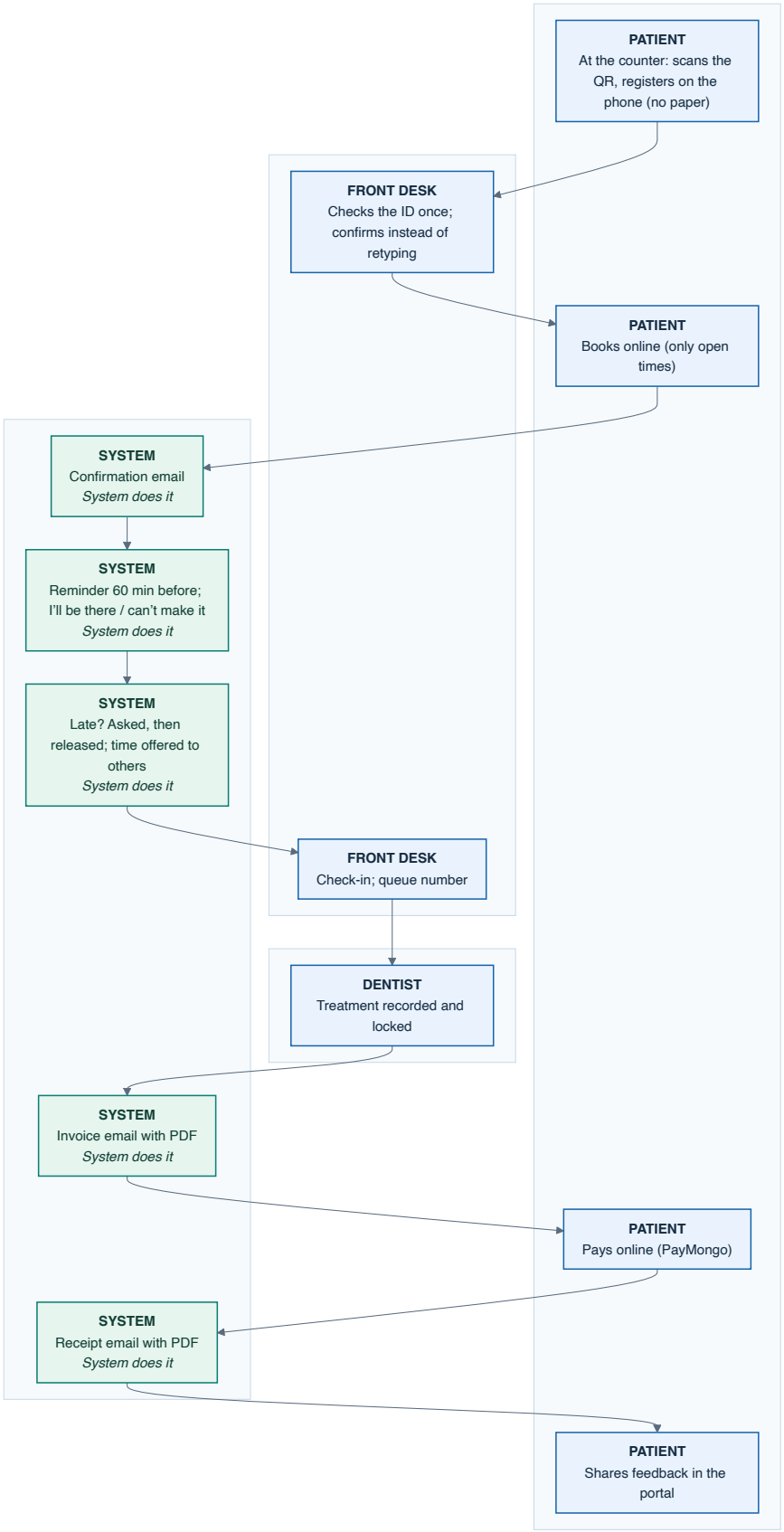


End-to-end patient journey: with the system (TO-BE)

Step from the interview (citation shown) · To confirm with the clinic · The system does it · A person does it in the system · Mock only · Planned (not built)

TO-BE (what the system does today)

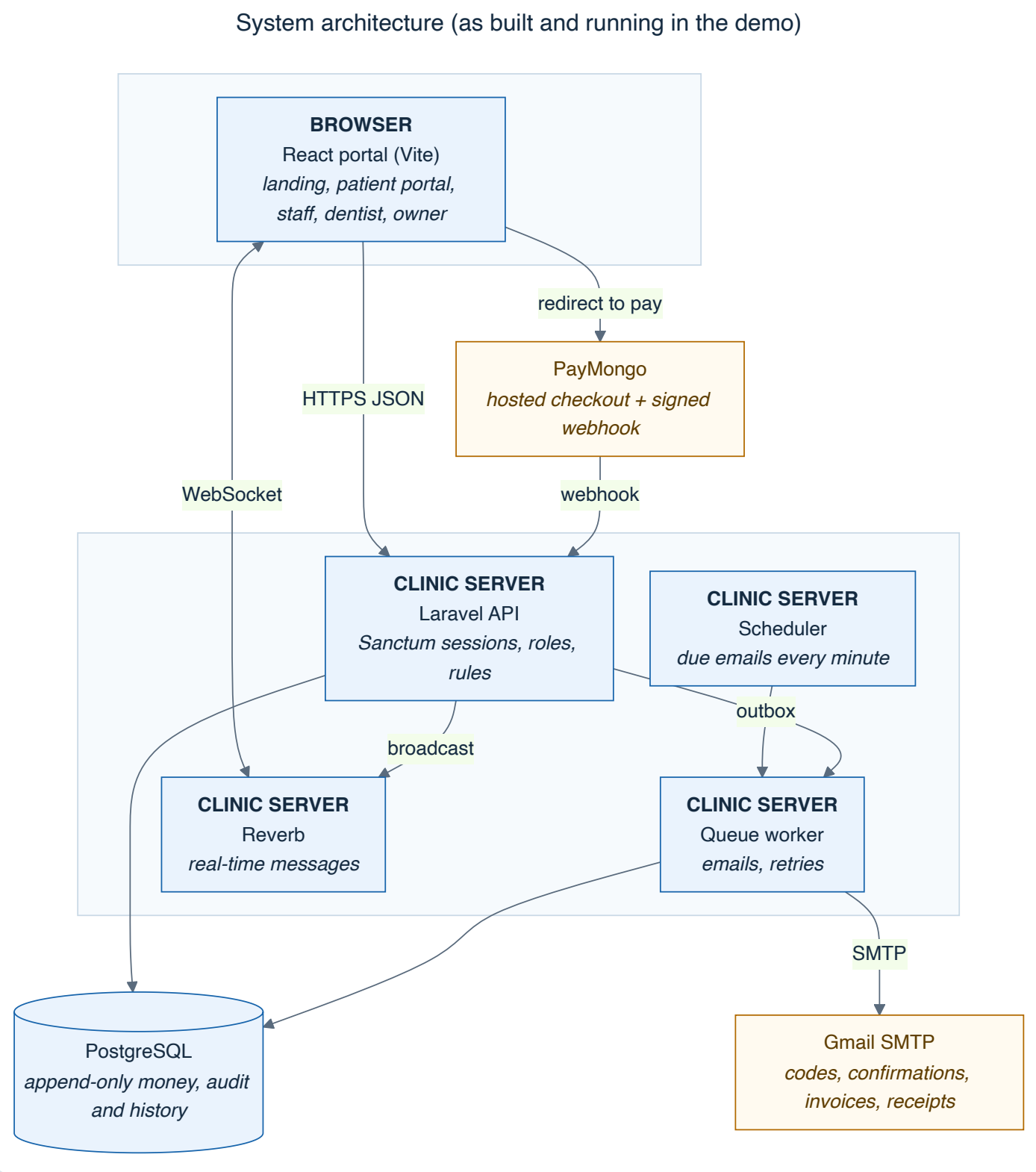
Patient journey: TO-BE (with the system)



System architecture (as built)

Running today in the demo: one clinic server, PostgreSQL, Gmail SMTP and PayMongo test mode.

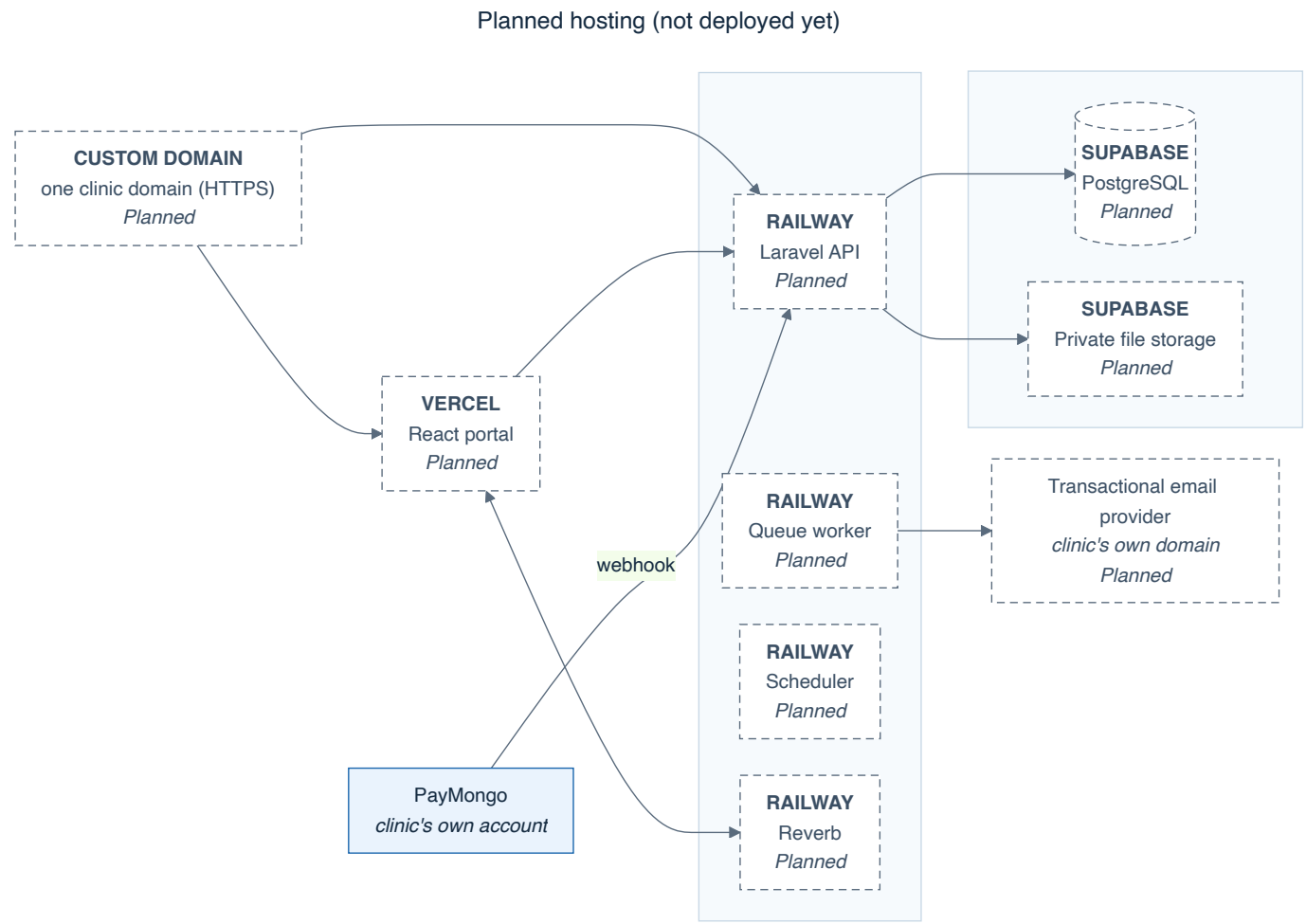
As built and running in the demo



Planned hosting

Not deployed. Dashed boxes are planned; PayMongo would use the clinic's own account.

Planned hosting (not deployed)



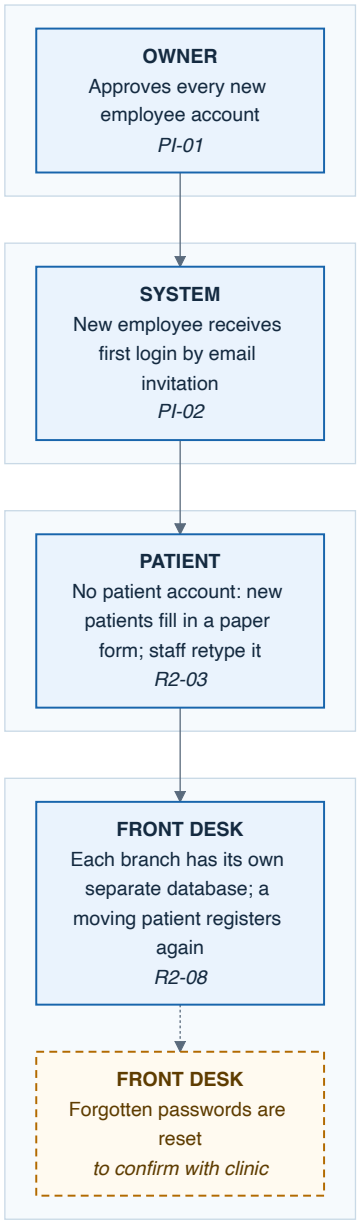
M1 User Management

Partial

AS-IS sources: PI-01, PI-02, R2-03, R2-08 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

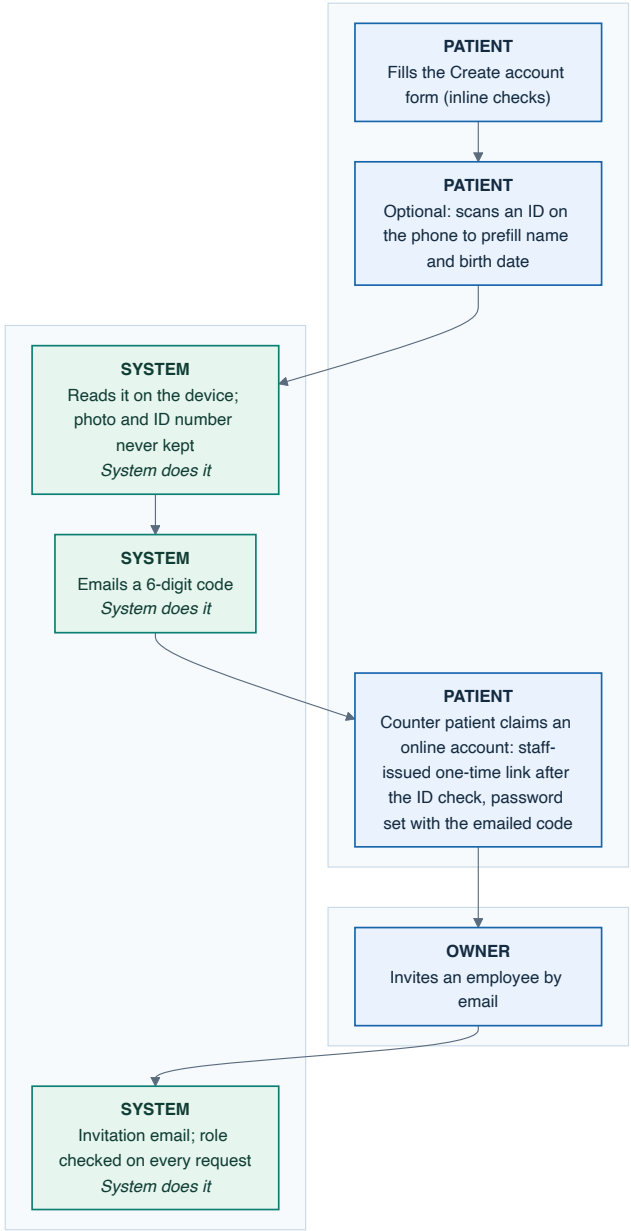
AS-IS (before)

M1 User Management: AS-IS (before)



TO-BE (with the system)

M1 User Management: TO-BE (with the system)



	Before	Now
Steps	5 known (1 to confirm)	7 in the system
Who does it	Patient, Front desk, Owner, System	Patient, Owner, System
Steps the system does by itself	—	3
Estimated time	About 15 minutes for a new patient's paper sign-in (R2-05)	To be measured in pilot
Errors removed	—	Accounts are created only after the email is proven; no shared logins; one record for all branches (R2-08)
Automation type	Manual	Event-driven (code and invitation emails), rule-based (role checks)

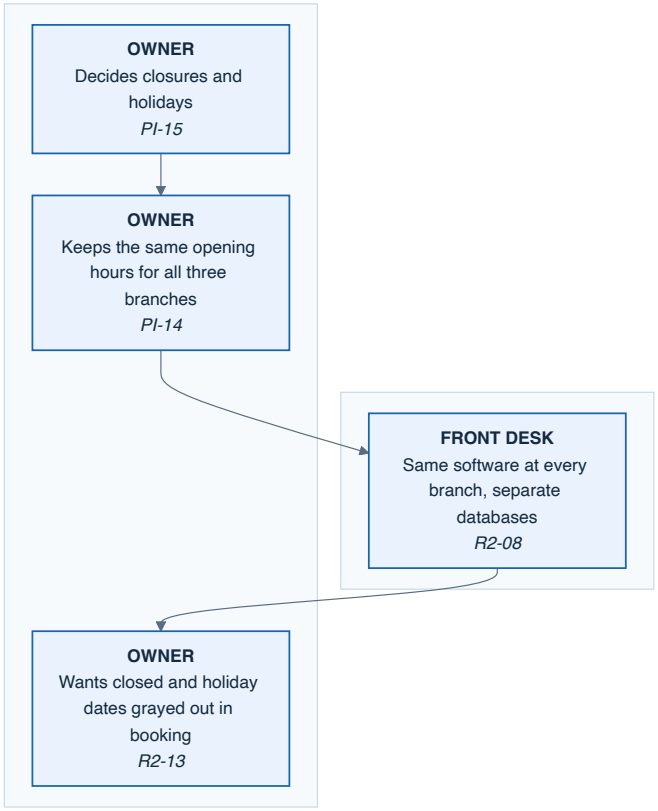
M2 Branch Management

Partial

AS-IS sources: PI-14, PI-15, R2-08, R2-13

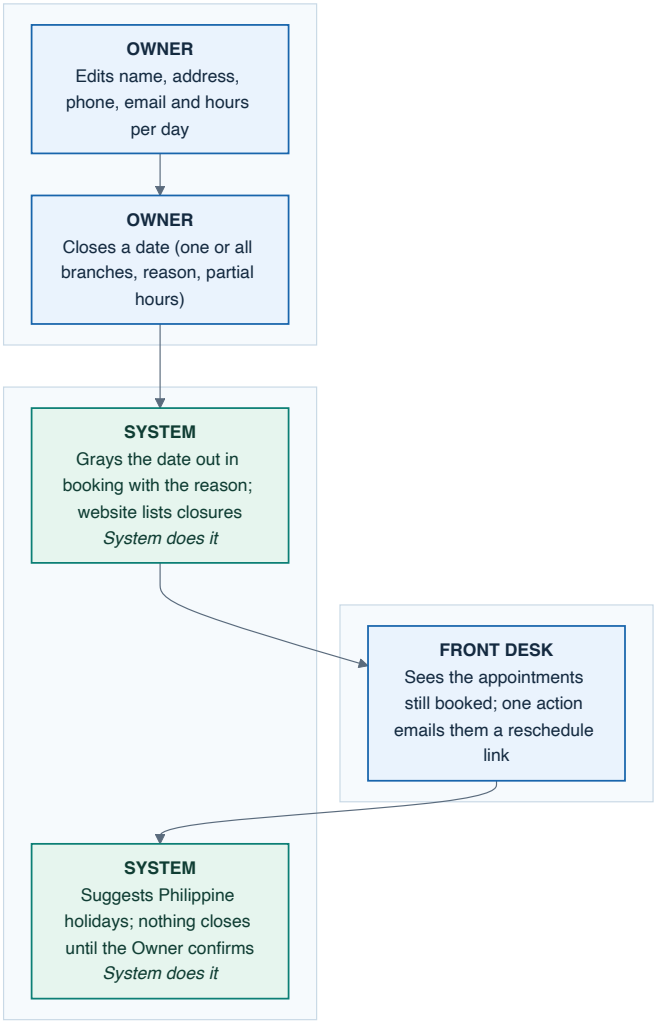
AS-IS (before)

M2 Branch Management: AS-IS (before)



TO-BE (with the system)

M2 Branch Management: TO-BE (with the system)



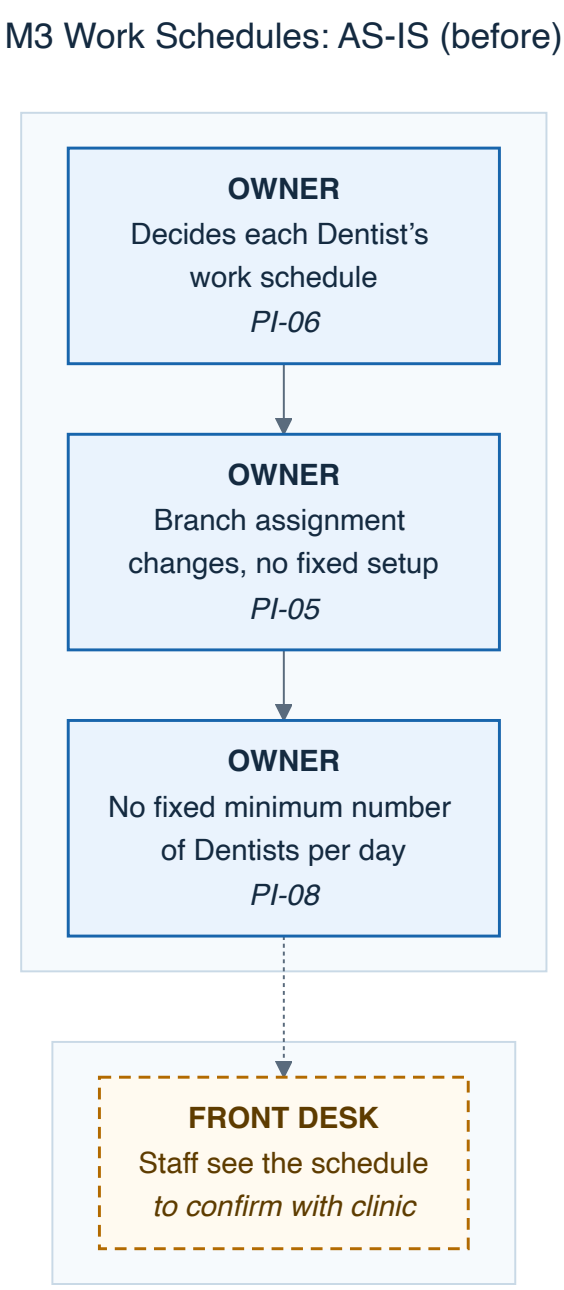
	Before	Now
Steps	4 known	5 in the system
Who does it	Front desk, Owner	Front desk, Owner, System
Steps the system does by itself	—	2
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Old or different hours in different places; bookings on closed days
Automation type	Manual	Rule-based

M3 Work Schedules

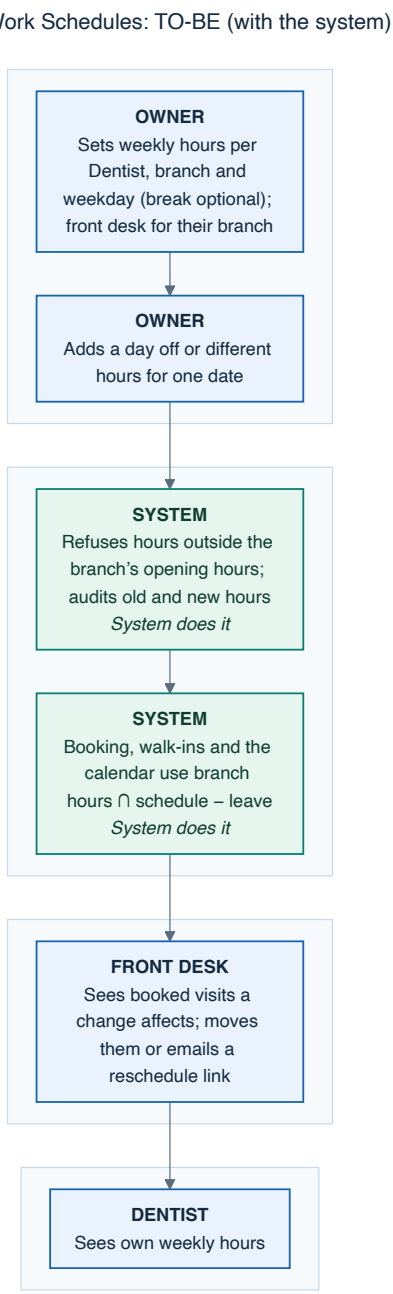
Implemented

AS-IS sources: PI-05, PI-06, PI-08 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

AS-IS (before)



TO-BE (with the system)



	Before	Now
Steps	4 known (1 to confirm)	6 in the system
Who does it	Front desk, Owner	Front desk, Dentist, Owner, System
Steps the system does by itself	—	2
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Booking outside a Dentist's hours; a change silently leaving patients without a Dentist
Automation type	Manual	Rule-based

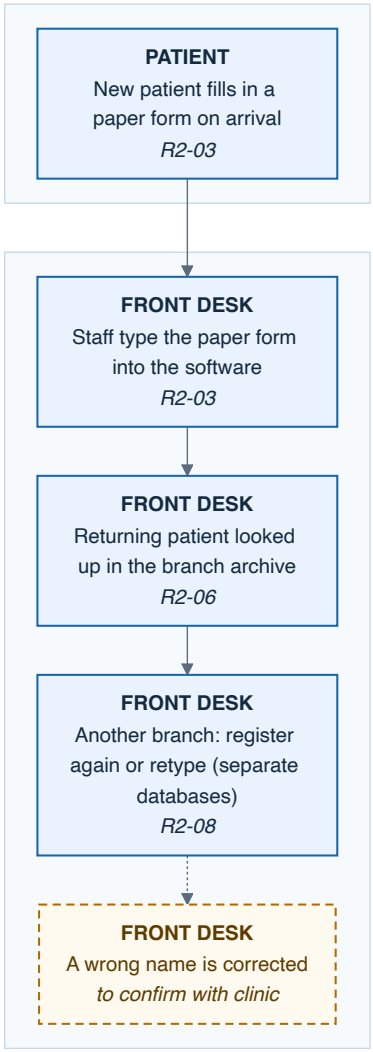
M4 Patient Records

Partial

AS-IS sources: R2-03, R2-06, R2-08 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

AS-IS (before)

M4 Patient Records: AS-IS (before)

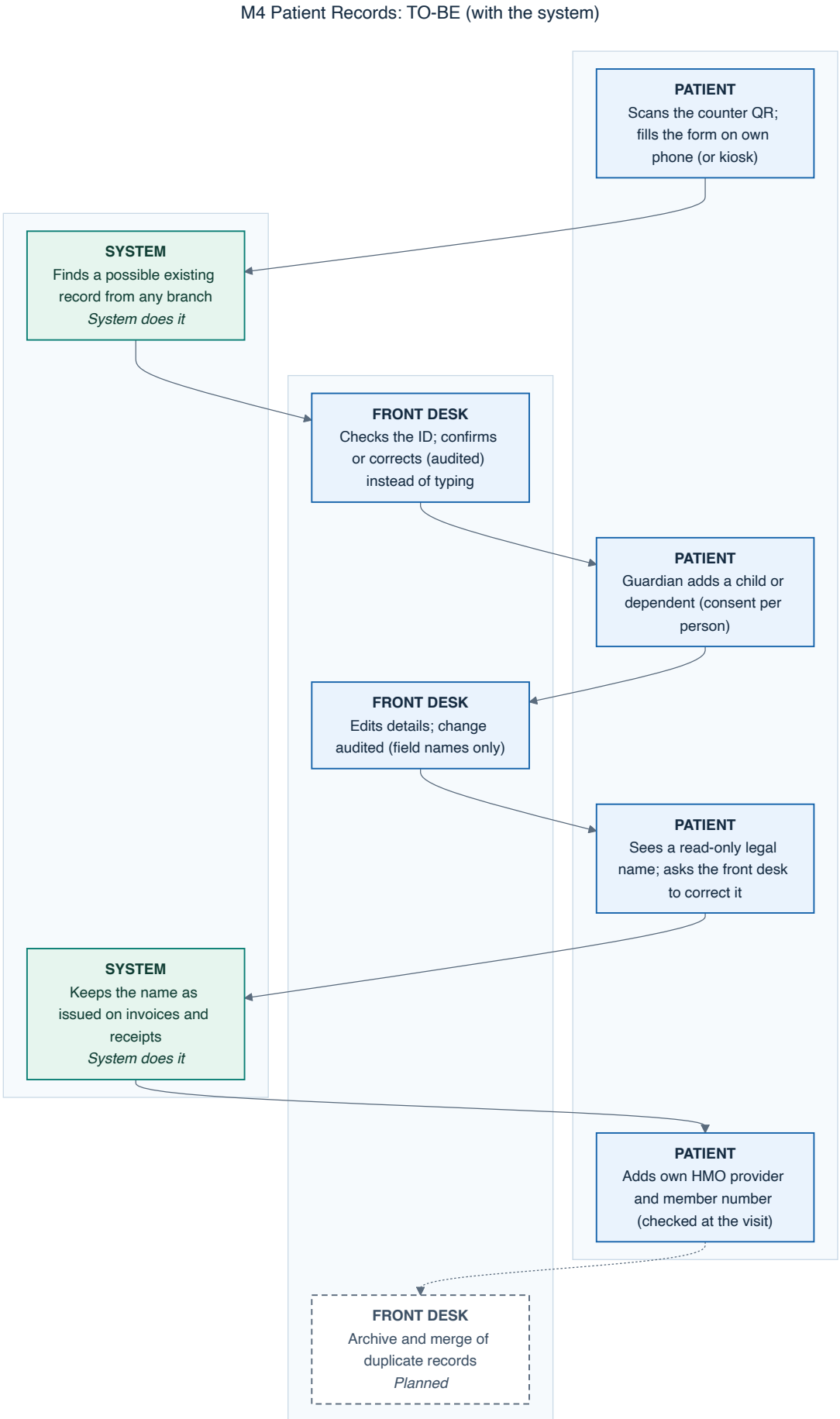


	Before	Now
Steps	5 known (1 to confirm)	8 in the system + 1 planned or mock
Who does it	Patient, Front desk	Patient, Front desk, System
Steps the system does by itself	—	2
Estimated time	About 15 minutes of paper sign-in, then retyping (R2-03, R2-05)	Phone self-registration plus a staff check: to be measured in pilot
Errors removed	—	Retyping paper forms (R2-03) and re-registering at another branch (R2-08); untracked edits
Automation type	Manual	Rule-based (audit, name snapshot)

M4 Patient Records: TO-BE

Partial

TO-BE (with the system)

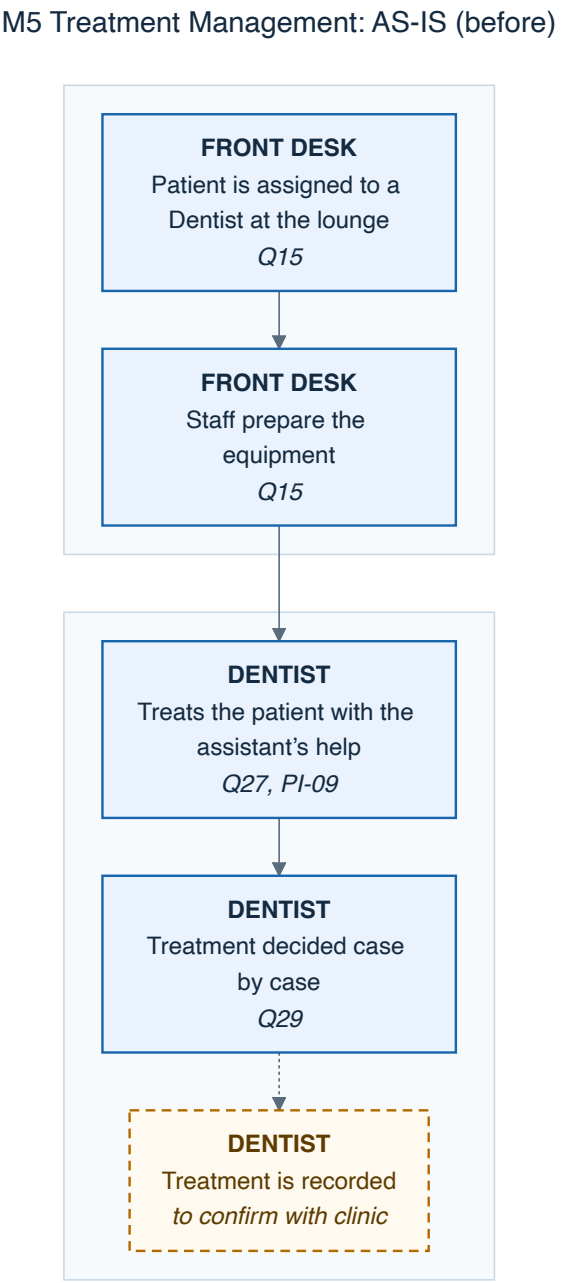


M5 Treatment Management

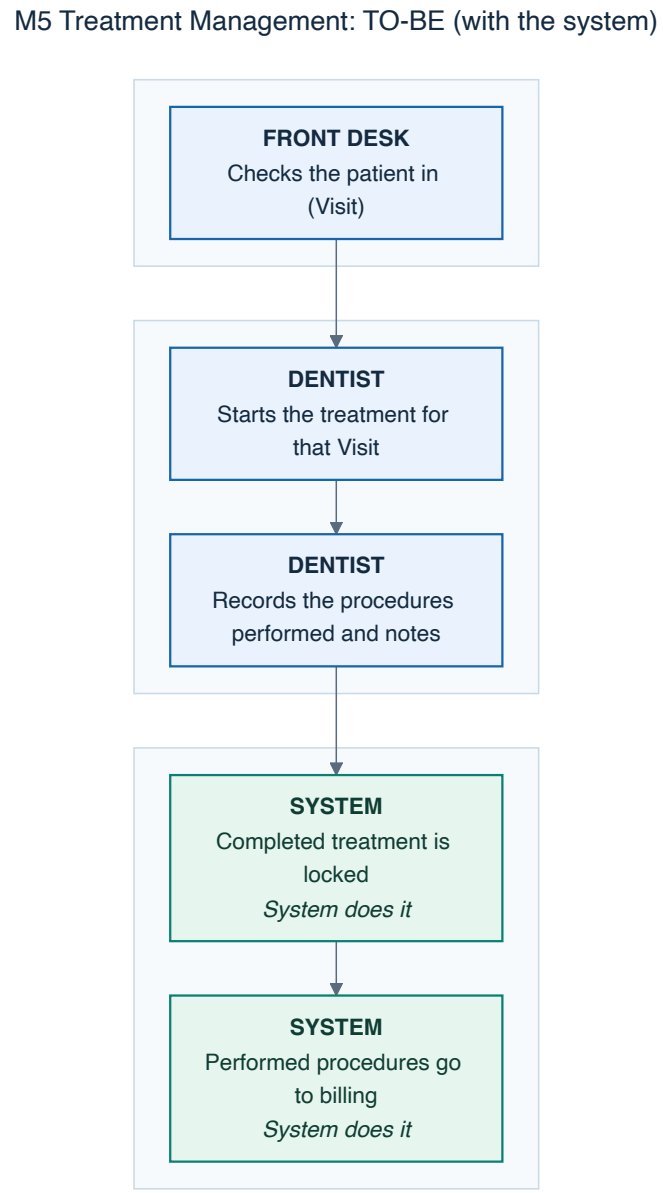
Implemented

AS-IS sources: Q15, Q27, PI-09, Q29 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

AS-IS (before)



TO-BE (with the system)



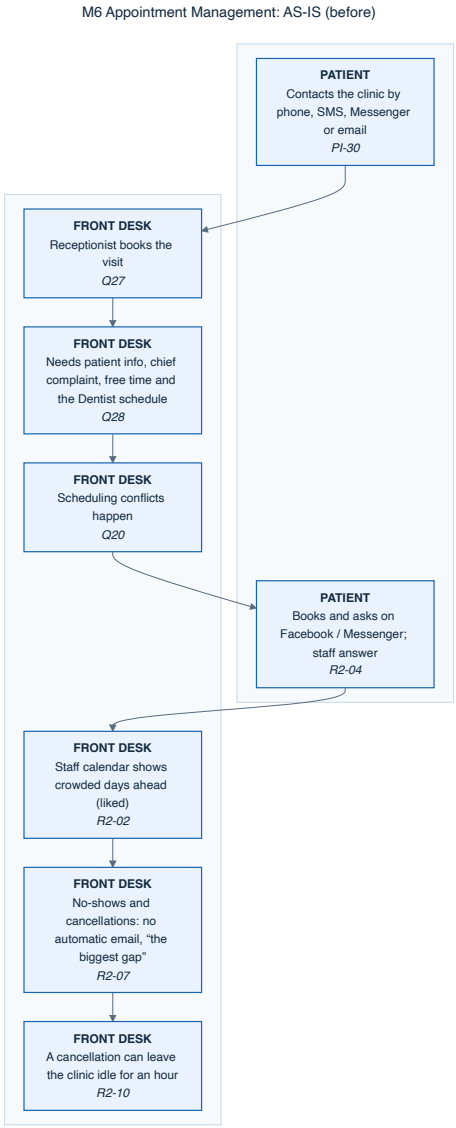
	Before	Now
Steps	5 known (1 to confirm)	5 in the system
Who does it	Front desk, Dentist	Front desk, Dentist, System
Steps the system does by itself	—	2
Estimated time	About 20–60 minutes per procedure, depending on the treatment (R2-05)	Unchanged; to be measured in pilot
Errors removed	—	Treatment changed after completion; charges not matching the work done
Automation type	Manual	Rule-based (lock), event-driven (to billing)

M6 Appointment Management

Implemented

AS-IS sources: PI-30, Q20, Q27, Q28, R2-02, R2-04, R2-07, R2-10

AS-IS (before)



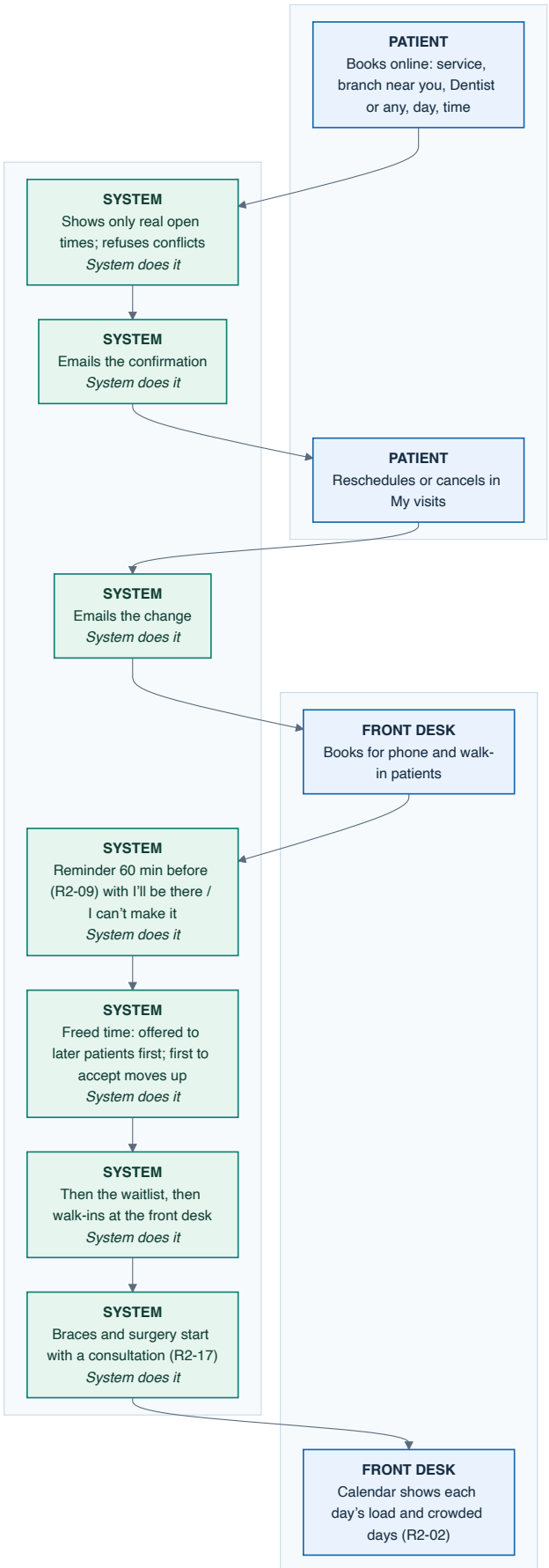
	Before	Now
Steps	8 known	11 in the system
Who does it	Patient, Front desk	Patient, Front desk, System
Steps the system does by itself	—	7
Estimated time	Not measured; idle hour after a cancellation (R2-10)	To be measured in pilot
Errors removed	—	Scheduling conflicts (Q20) blocked by the database; no-shows (R2-07) met by reminders, one-click answers and freed-time offers
Automation type	Manual	Rule-based (open times, no overlaps), event-driven (emails), scheduled (reminders, waitlist)

M6 Appointment Management: TO-BE

Implemented

TO-BE (with the system)

M6 Appointment Management: TO-BE (with the system)

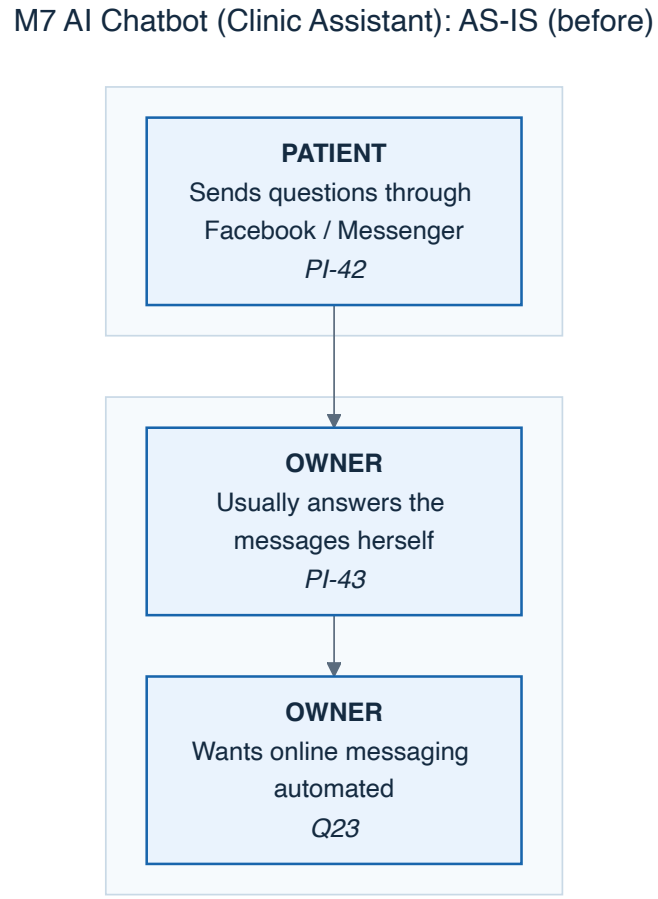


M7 AI Chatbot (Clinic Assistant)

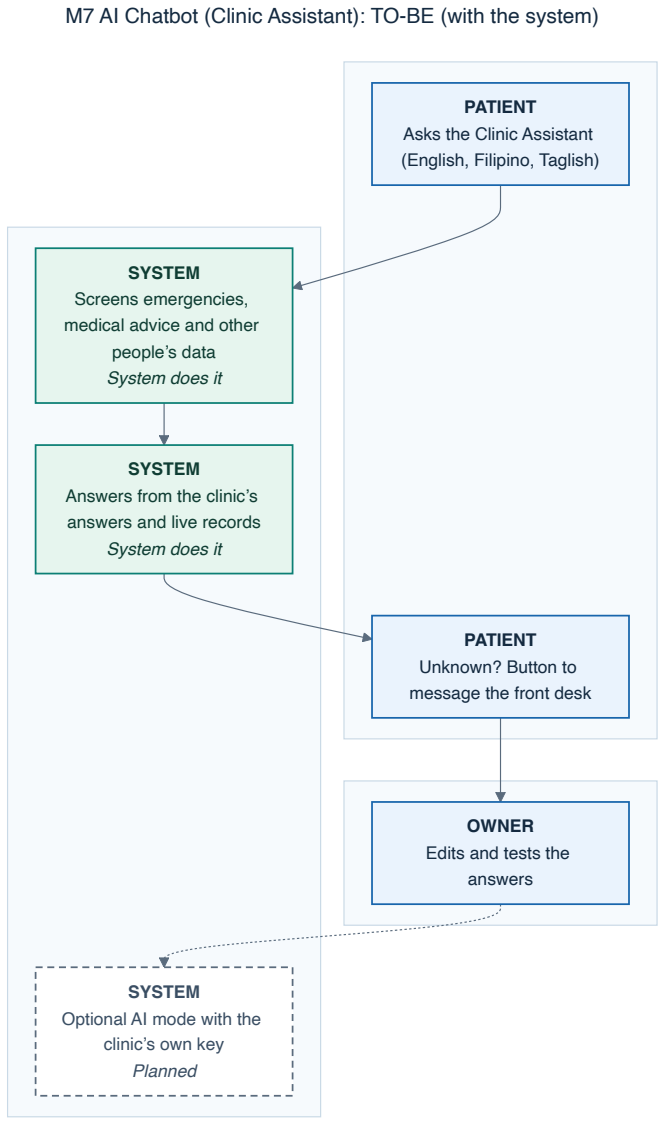
Implemented

AS-IS sources: PI-42, PI-43, Q23

AS-IS (before)



TO-BE (with the system)



	Before	Now
Steps	3 known	5 in the system + 1 planned or mock
Who does it	Patient, Owner	Patient, Owner, System
Steps the system does by itself	—	2
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Routine questions no longer wait for the Owner
Automation type	Manual	Rule-based (knowledge matching)

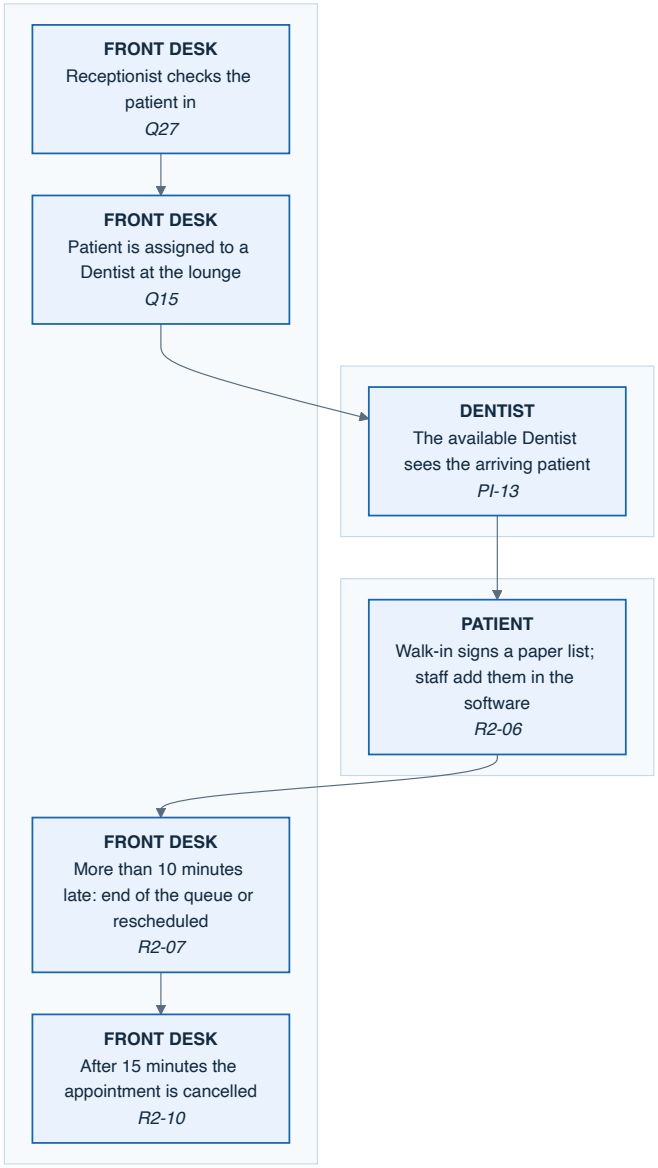
M8 Patient Visits

Partial

AS-IS sources: PI-13, Q15, Q27, R2-06, R2-07, R2-10

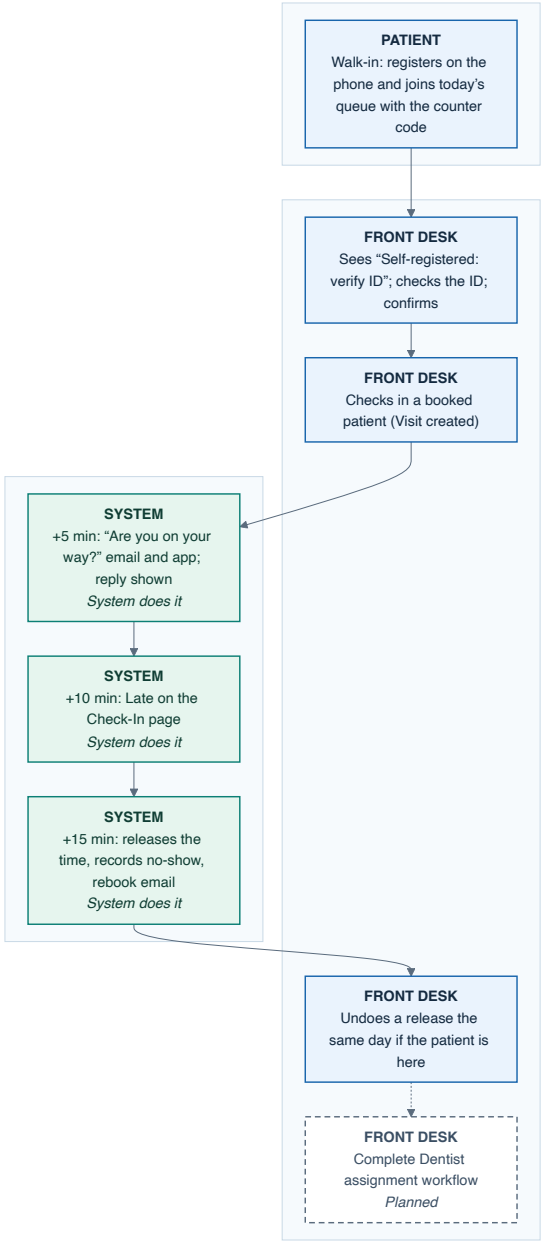
AS-IS (before)

M8 Patient Visits: AS-IS (before)



TO-BE (with the system)

M8 Patient Visits: TO-BE (with the system)



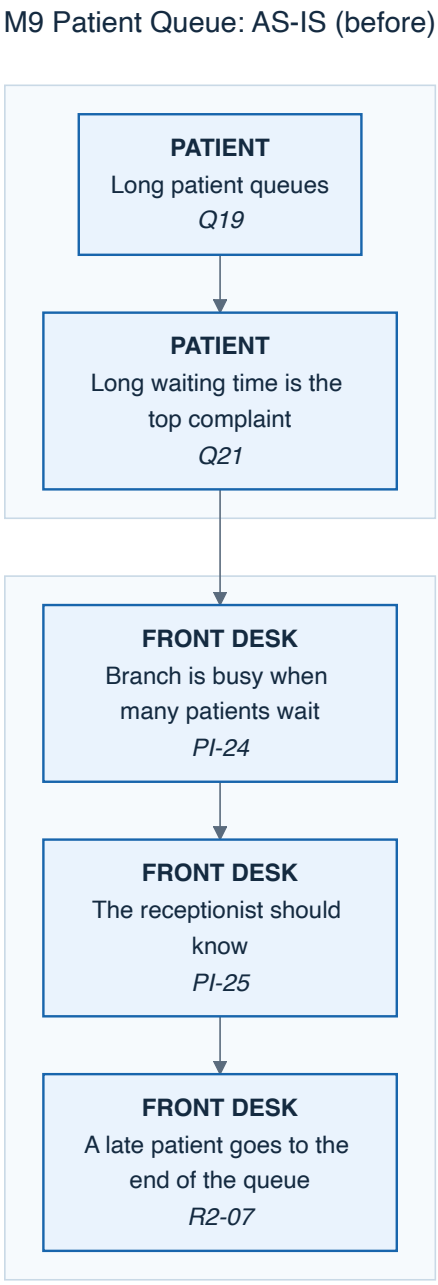
	Before	Now
Steps	6 known	7 in the system + 1 planned or mock
Who does it	Patient, Front desk, Dentist	Patient, Front desk, System
Steps the system does by itself	—	3
Estimated time	About 15 minutes for a new patient's sign-in (R2-05)	Phone self-registration plus a staff check: to be measured in pilot
Errors removed	—	Arrivals not linked to their booking; missed visits noticed late
Automation type	Manual	Rule-based, scheduled (no-show flag; a person decides)

M9 Patient Queue

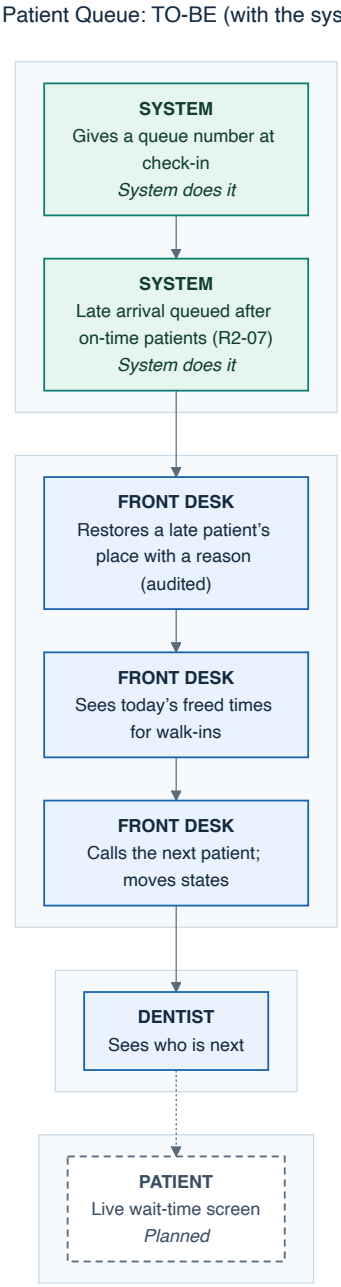
Partial

AS-IS sources: PI-24, PI-25, Q19, Q21, R2-07

AS-IS (before)



TO-BE (with the system)



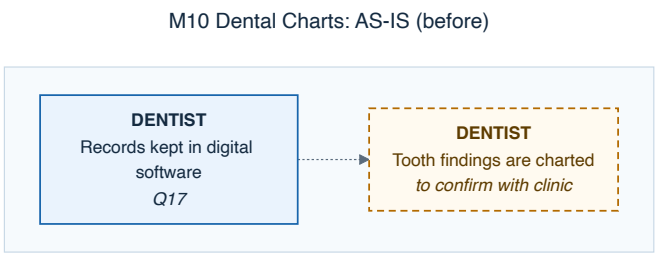
	Before	Now
Steps	5 known	6 in the system + 1 planned or mock
Who does it	Patient, Front desk	Front desk, Dentist, System
Steps the system does by itself	—	2
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Not measured
Automation type	Manual	Rule-based (numbering)

M10 Dental Charts

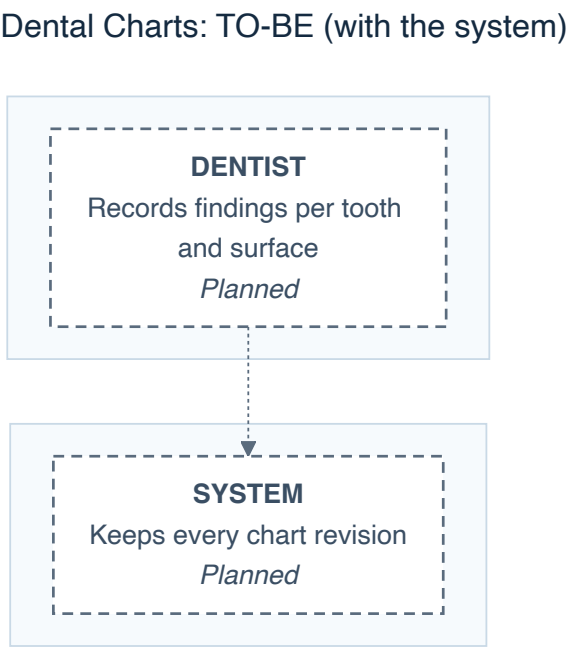
Planned

AS-IS sources: Q17 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

AS-IS (before)



TO-BE (with the system)



	Before	Now
Steps	2 known (1 to confirm)	0 in the system + 2 planned or mock
Who does it	Dentist	Nobody yet (planned)
Steps the system does by itself	—	0
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Not built yet
Automation type	Manual	Planned

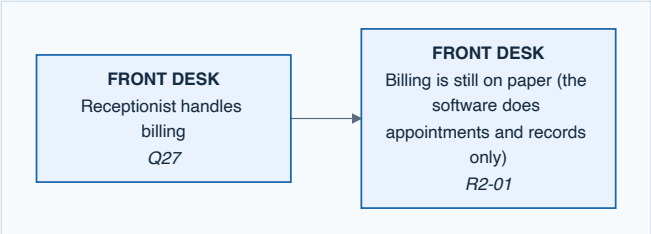
M11 Billing Management

Implemented

AS-IS sources: Q27, R2-01

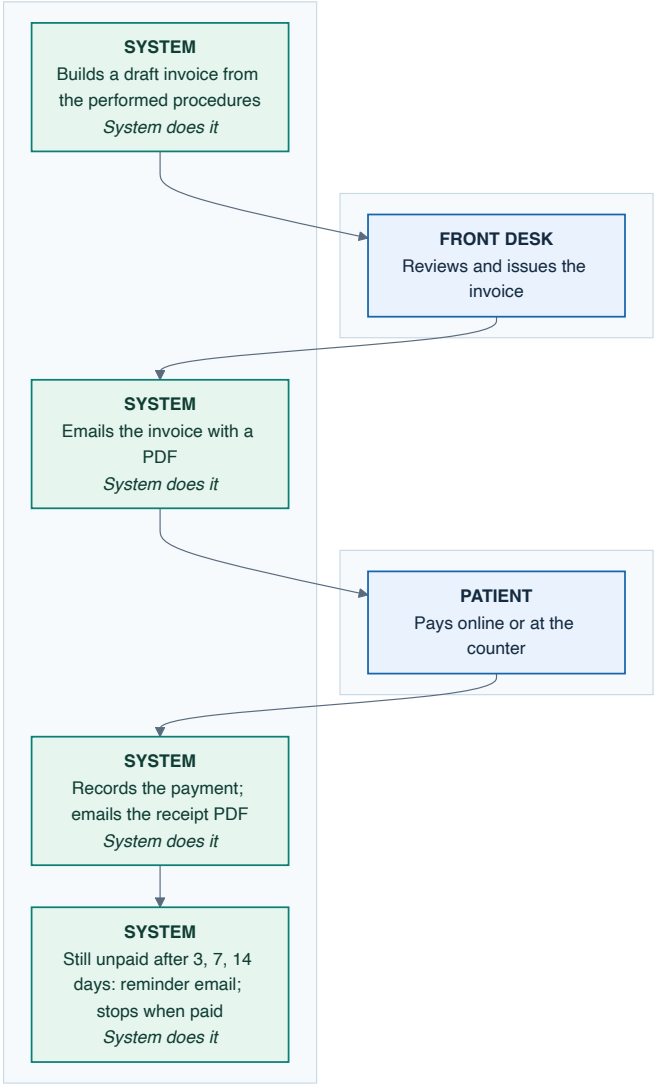
AS-IS (before)

M11 Billing Management: AS-IS (before)



TO-BE (with the system)

M11 Billing Management: TO-BE (with the system)



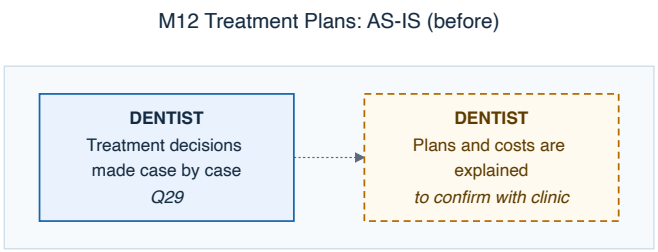
	Before	Now
Steps	2 known	6 in the system
Who does it	Front desk	Patient, Front desk, System
Steps the system does by itself	—	4
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Charges not matching the treatment; edited receipts (database blocks changes); unpaid balances forgotten
Automation type	Manual	Event-driven (draft, emails), rule-based (immutable charges), scheduled (overdue reminders)

M12 Treatment Plans

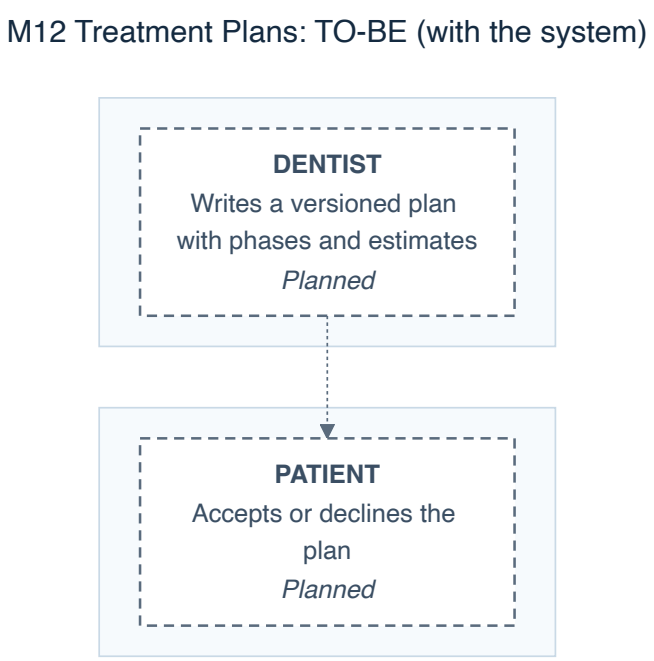
Planned

AS-IS sources: Q29 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

AS-IS (before)



TO-BE (with the system)



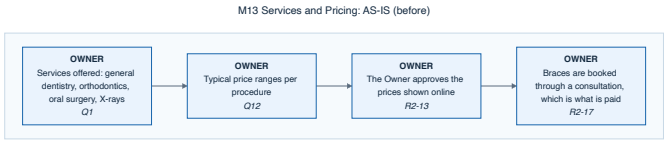
	Before	Now
Steps	2 known (1 to confirm)	0 in the system + 2 planned or mock
Who does it	Dentist	Nobody yet (planned)
Steps the system does by itself	—	0
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Not built yet
Automation type	Manual	Planned

M13 Services and Pricing

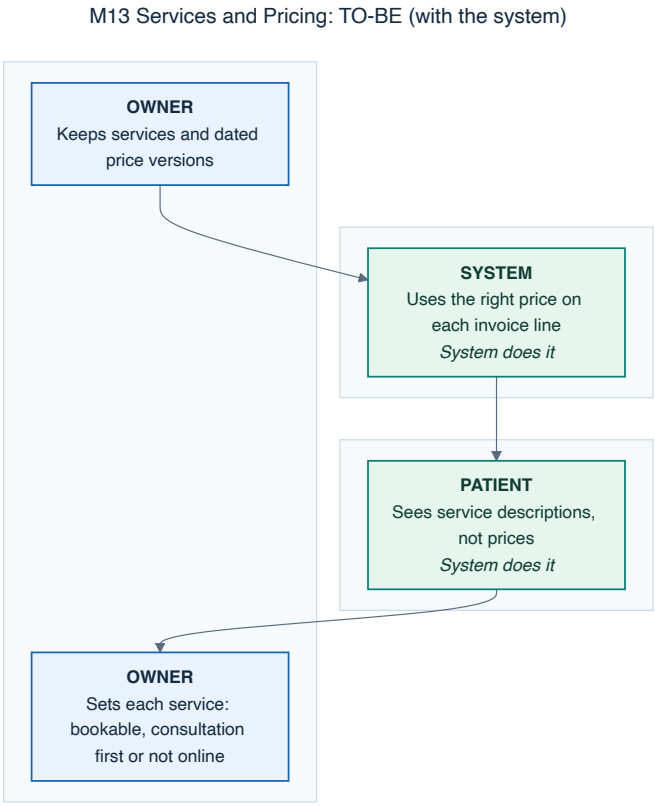
Partial

AS-IS sources: Q1, Q12, R2-13, R2-17

AS-IS (before)



TO-BE (with the system)



	Before	Now
Steps	4 known	4 in the system
Who does it	Owner	Patient, Owner, System
Steps the system does by itself	—	2
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Wrong or outdated prices on invoices
Automation type	Manual	Rule-based (effective price)

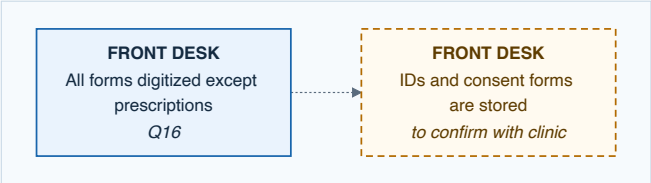
M14 Document Management

Partial

AS-IS sources: Q16 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

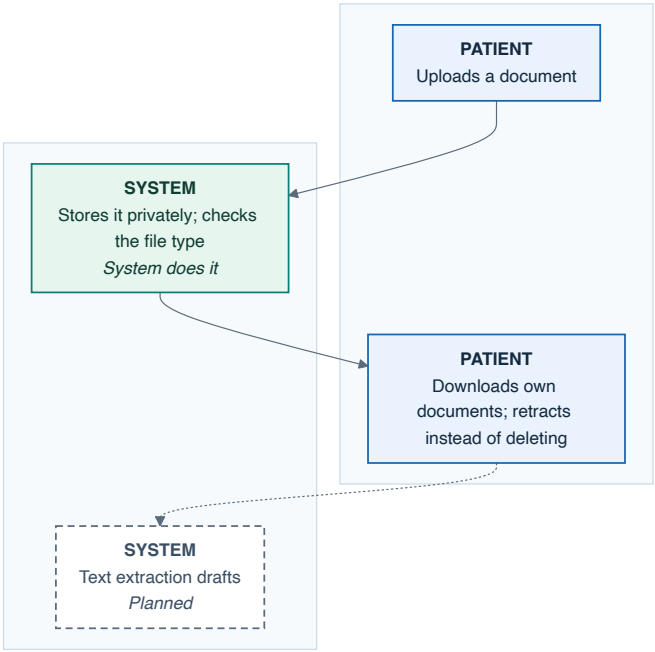
AS-IS (before)

M14 Document Management: AS-IS (before)



TO-BE (with the system)

M14 Document Management: TO-BE (with the system)



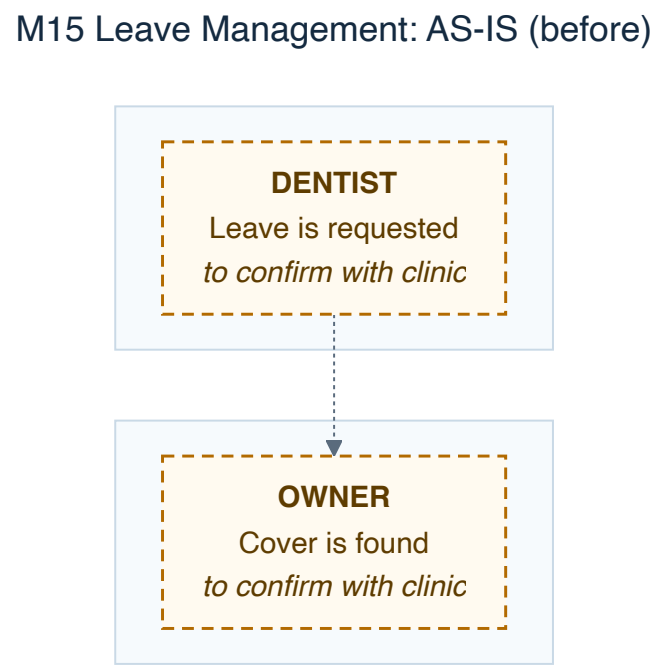
	Before	Now
Steps	2 known (1 to confirm)	3 in the system + 1 planned or mock
Who does it	Front desk	Patient, System
Steps the system does by itself	—	1
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Lost or shared files
Automation type	Manual	Rule-based

M15 Leave Management

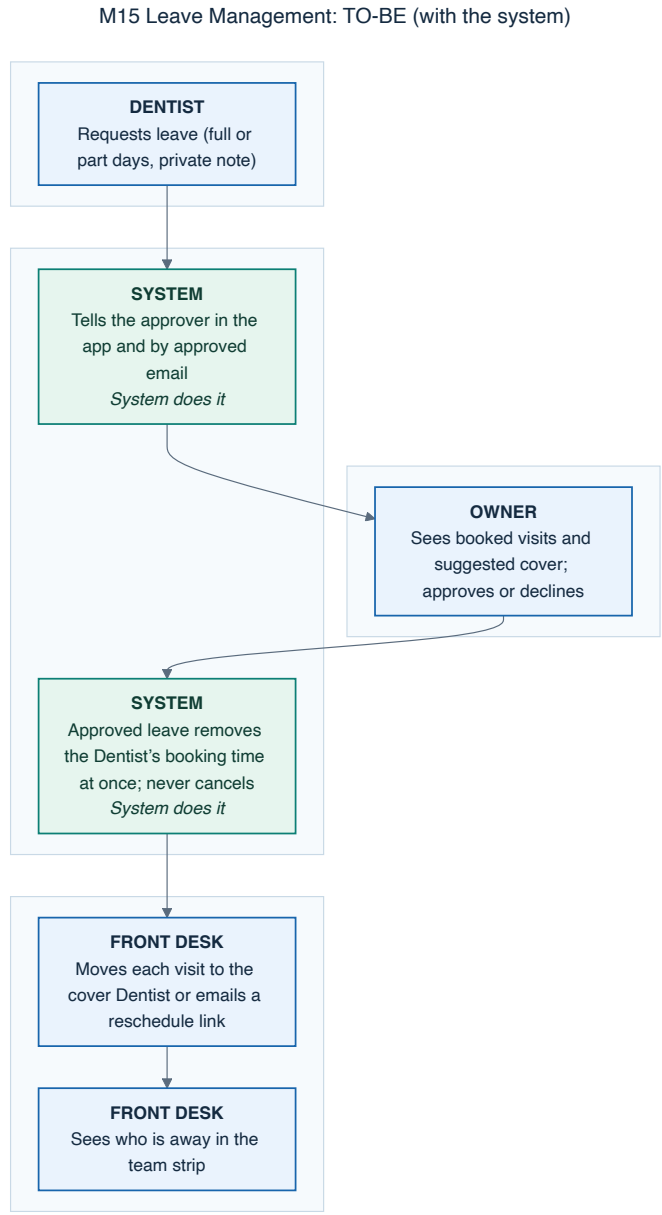
Implemented

AS-IS sources: none in the interview · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

AS-IS (before)



TO-BE (with the system)



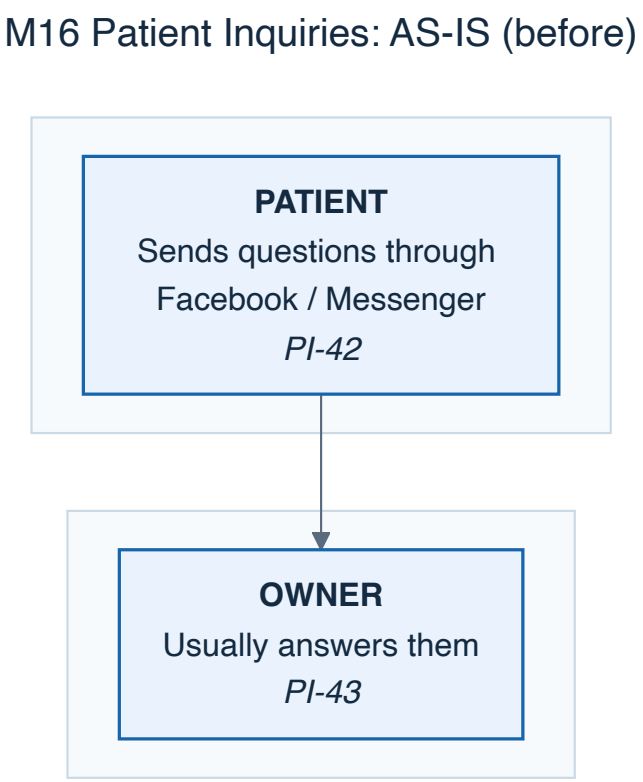
	Before	Now
Steps	2 known (2 to confirm)	6 in the system
Who does it	Dentist, Owner	Front desk, Dentist, Owner, System
Steps the system does by itself	—	2
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Patients booked with an absent Dentist
Automation type	Manual	Rule-based

M16 Patient Inquiries

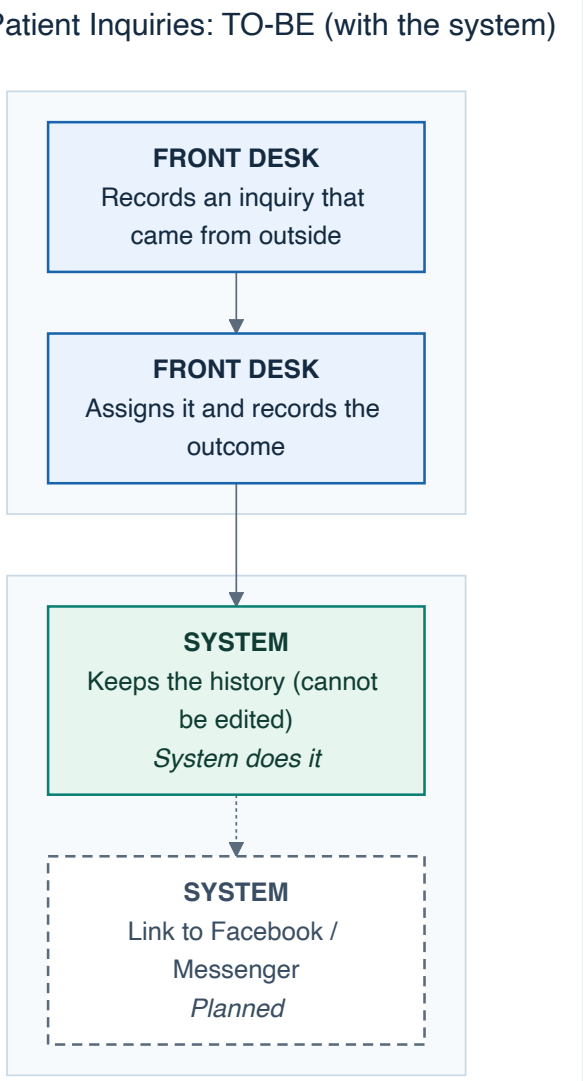
Partial

AS-IS sources: PI-42, PI-43

AS-IS (before)



TO-BE (with the system)



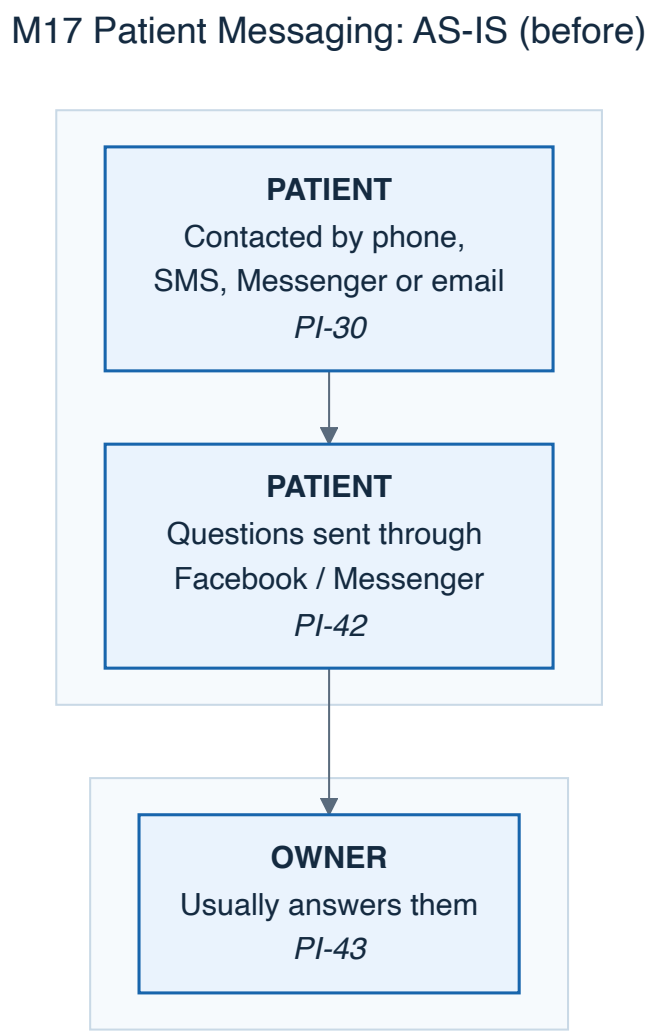
	Before	Now
Steps	2 known	3 in the system + 1 planned or mock
Who does it	Patient, Owner	Front desk, System
Steps the system does by itself	—	1
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Inquiries forgotten or answered twice
Automation type	Manual	Rule-based

M17 Patient Messaging

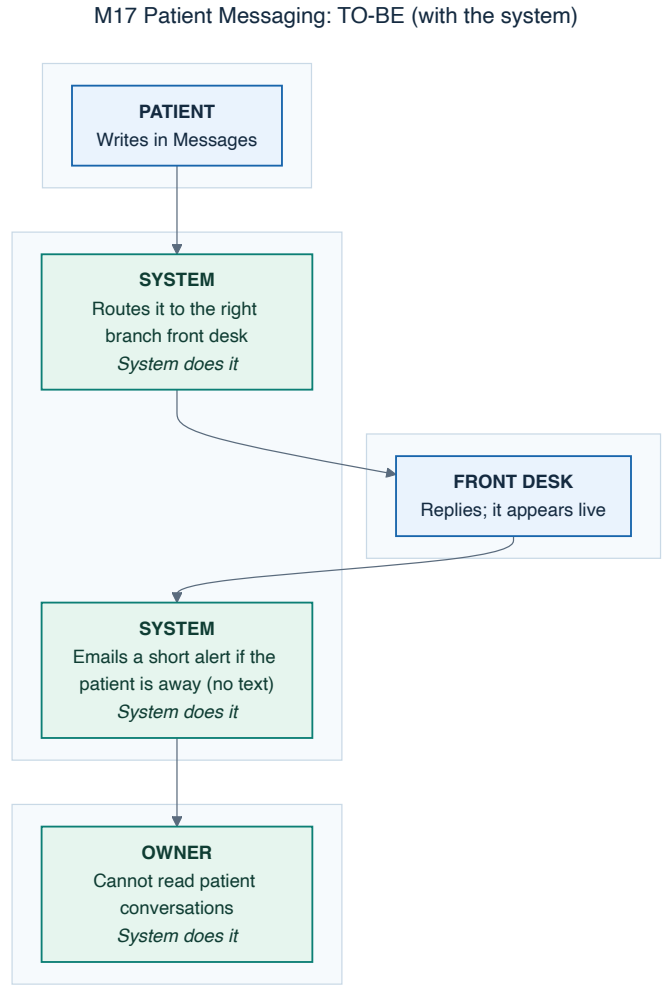
Implemented

AS-IS sources: PI-30, PI-42, PI-43

AS-IS (before)



TO-BE (with the system)



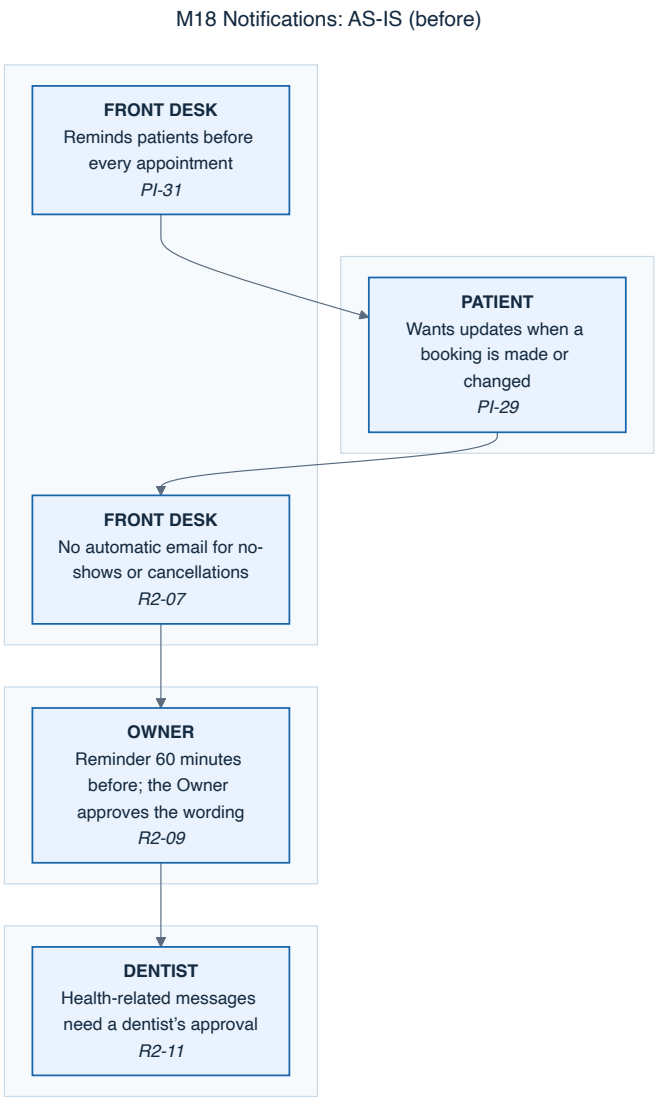
	Before	Now
Steps	3 known	5 in the system
Who does it	Patient, Owner	Patient, Front desk, Owner, System
Steps the system does by itself	—	3
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Messages to the wrong person or lost in personal accounts
Automation type	Manual	Event-driven (real-time, alert email)

M18 Notifications

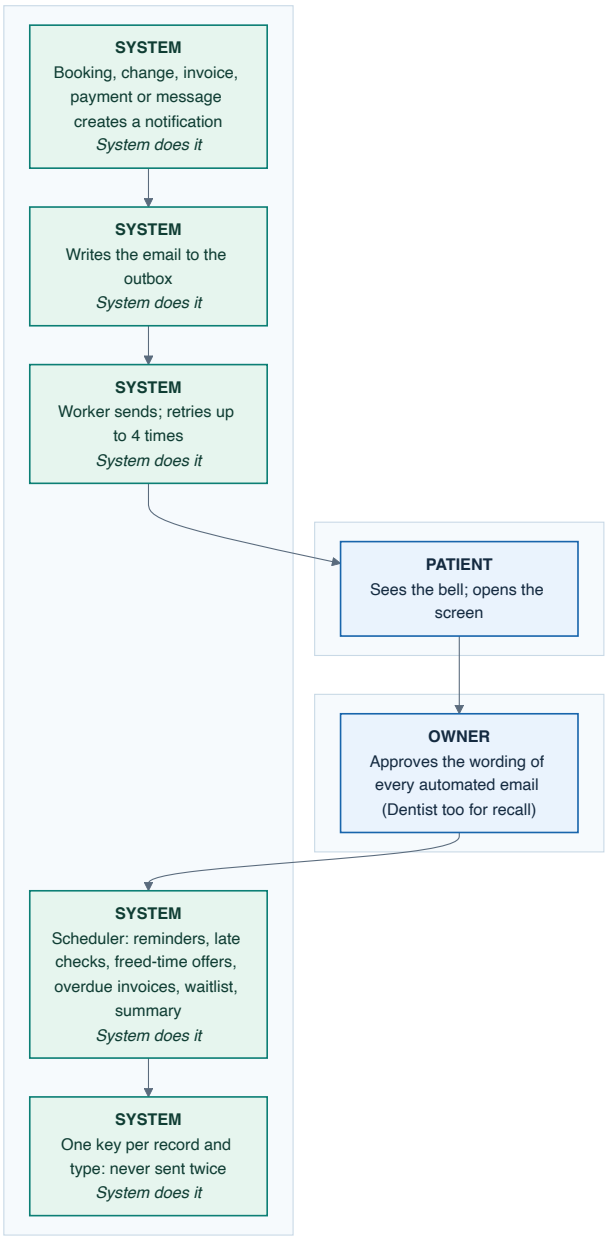
Implemented

AS-IS sources: PI-29, PI-31, R2-07, R2-09, R2-11

AS-IS (before)



TO-BE (with the system)



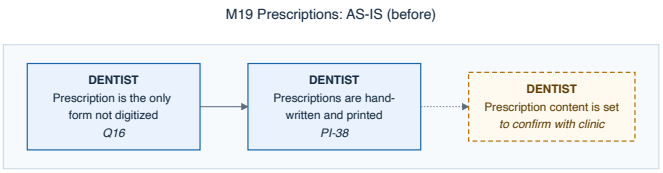
	Before	Now
Steps	5 known	7 in the system
Who does it	Patient, Front desk, Dentist, Owner	Patient, Owner, System
Steps the system does by itself	—	5
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Missed updates and reminders (PI-31); failed emails are retried; no duplicates
Automation type	Manual	Event-driven, scheduled (reminders, retries)

M19 Prescriptions

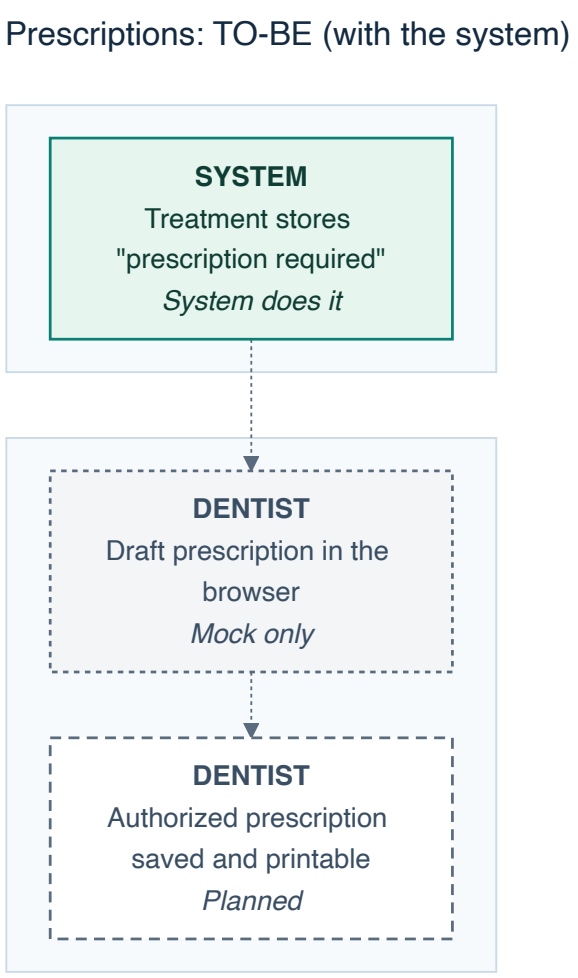
Mock only

AS-IS sources: PI-38, Q16 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

AS-IS (before)



TO-BE (with the system)



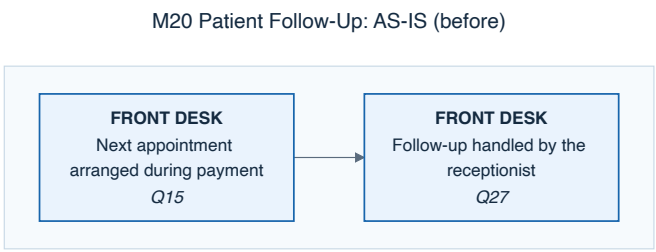
	Before	Now
Steps	3 known (1 to confirm)	1 in the system + 2 planned or mock
Who does it	Dentist	System
Steps the system does by itself	—	1
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Not built on the server yet
Automation type	Manual	Planned

M20 Patient Follow-Up

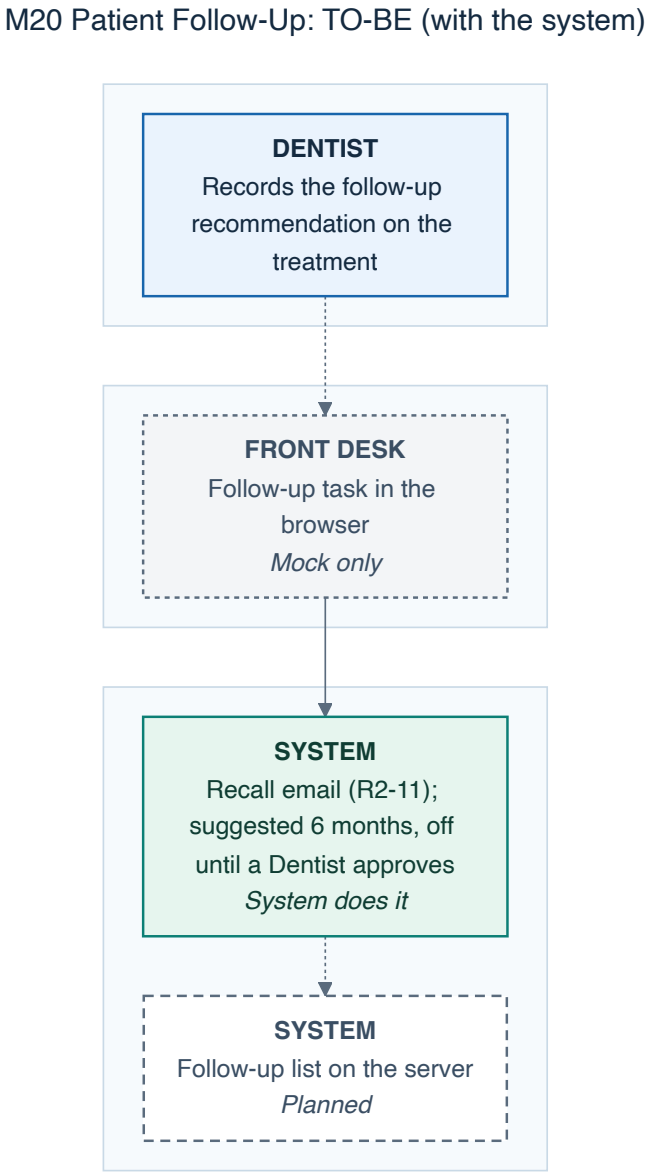
Mock only

AS-IS sources: Q15, Q27

AS-IS (before)



TO-BE (with the system)



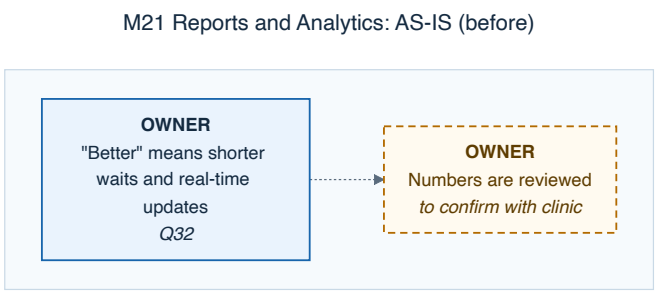
	Before	Now
Steps	2 known	2 in the system + 2 planned or mock
Who does it	Front desk	Dentist, System
Steps the system does by itself	—	1
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Follow-up tasks not on the server yet
Automation type	Manual	Scheduled recall email (off by default); follow-up tasks planned

M21 Reports and Analytics

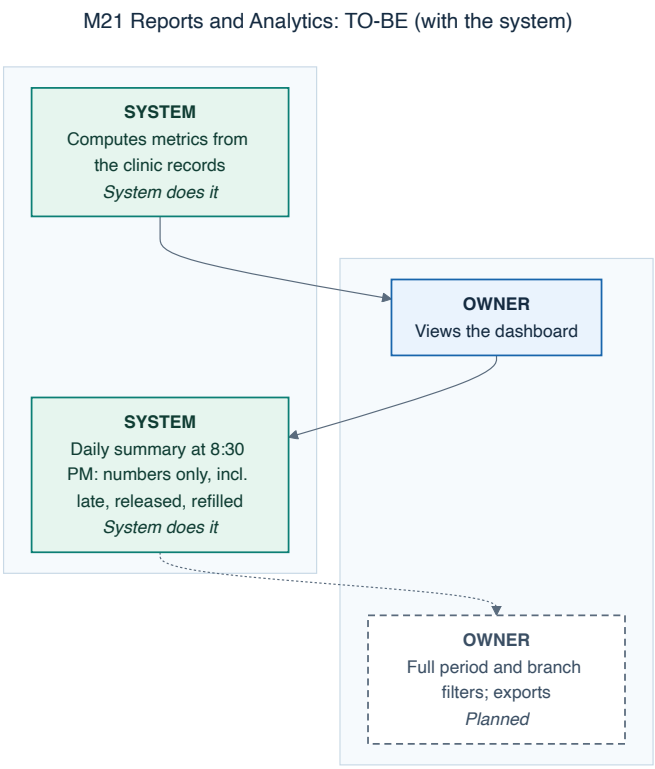
Partial

AS-IS sources: Q32 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

AS-IS (before)



TO-BE (with the system)



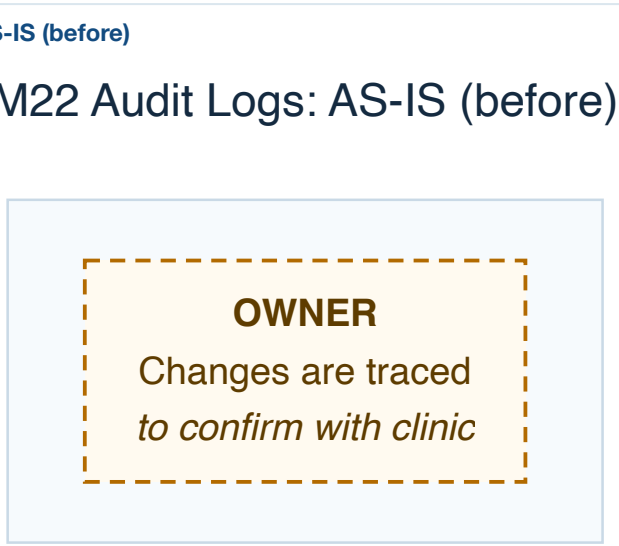
	Before	Now
Steps	2 known (1 to confirm)	3 in the system + 1 planned or mock
Who does it	Owner	Owner, System
Steps the system does by itself	—	2
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Numbers compiled by hand
Automation type	Manual	Rule-based

M22 Audit Logs

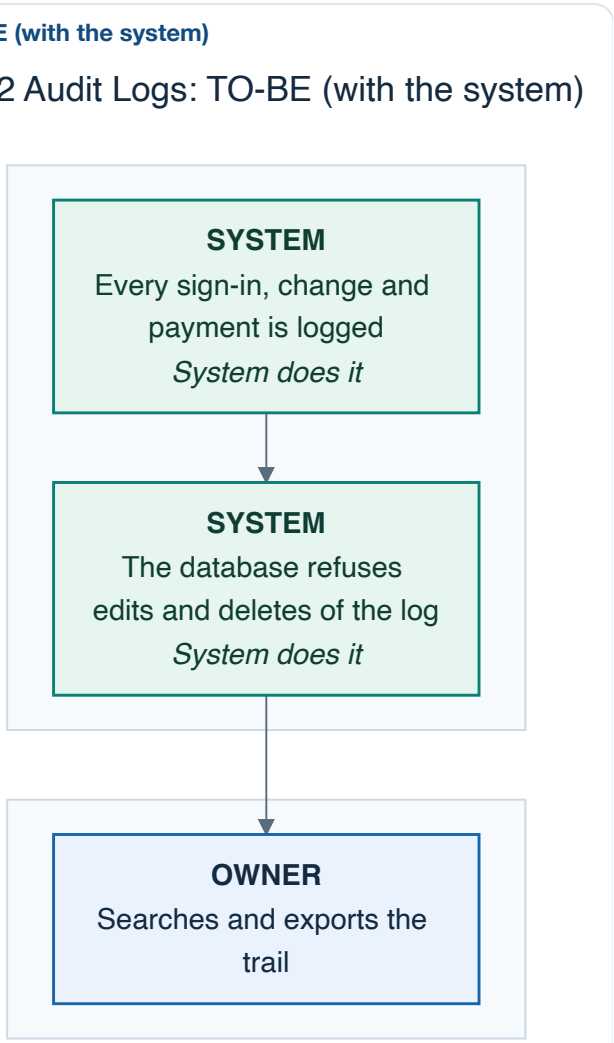
Implemented

AS-IS sources: none in the interview · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

AS-IS (before)



TO-BE (with the system)



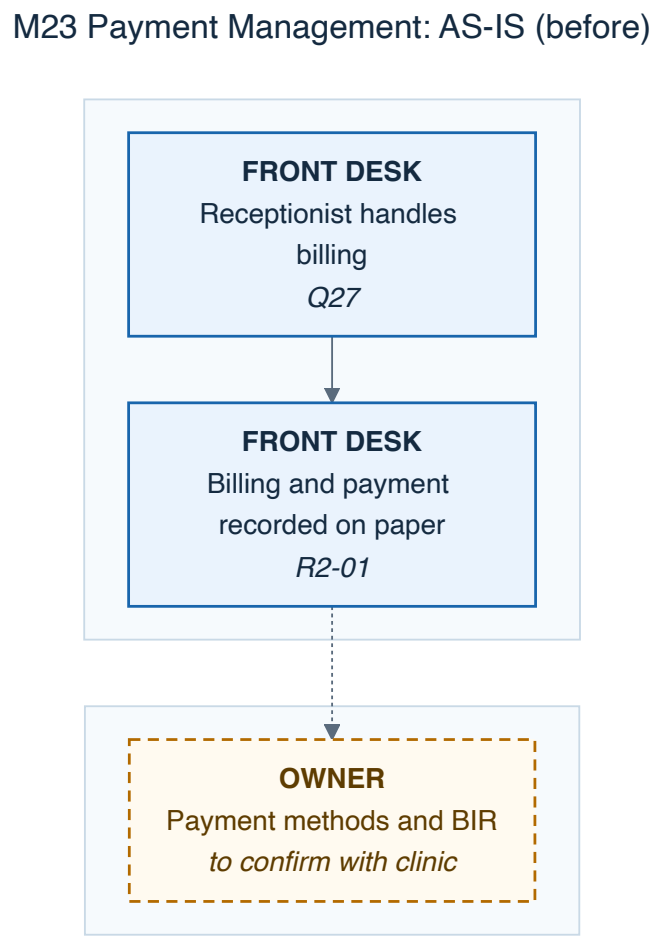
	Before	Now
Steps	1 known (1 to confirm)	3 in the system
Who does it	Owner	Owner, System
Steps the system does by itself	—	2
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Untraceable changes
Automation type	Manual	Event-driven

M23 Payment Management

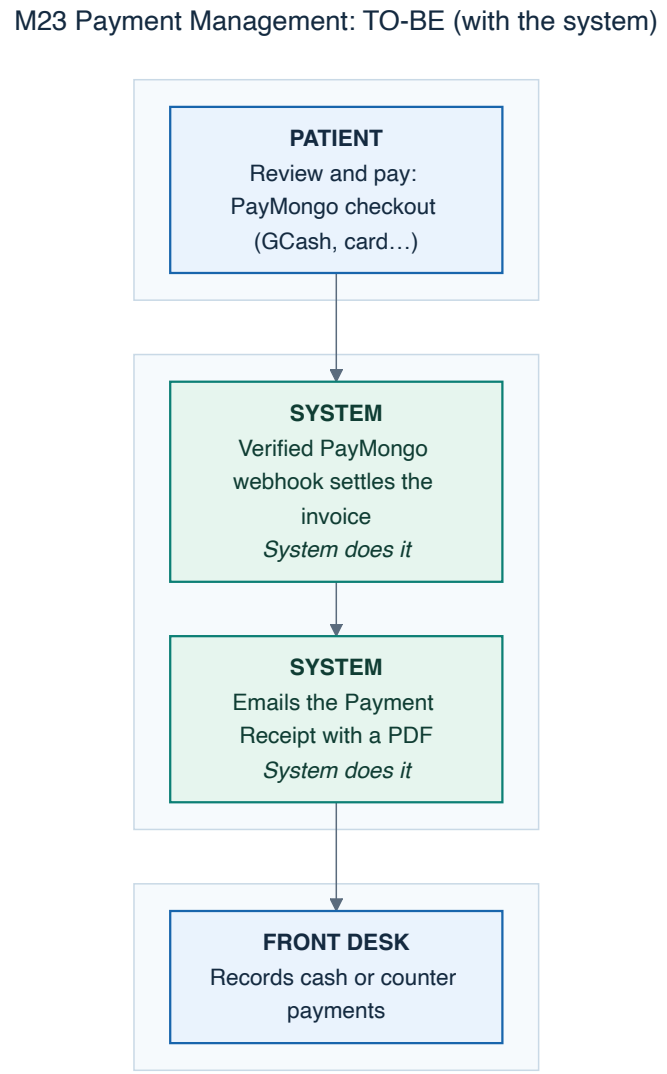
Implemented

AS-IS sources: Q27, R2-01 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

AS-IS (before)



TO-BE (with the system)



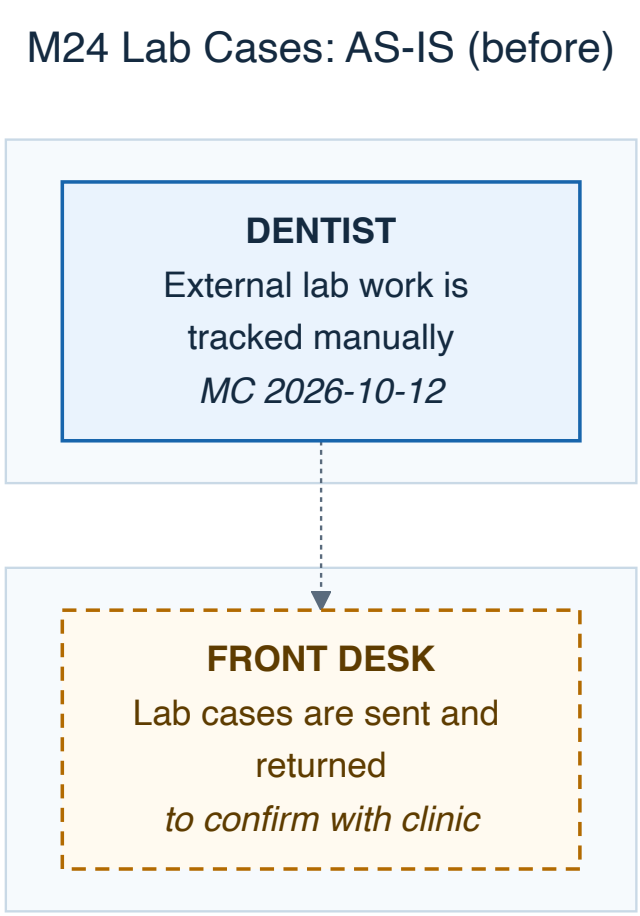
	Before	Now
Steps	3 known (1 to confirm)	4 in the system
Who does it	Front desk, Owner	Patient, Front desk, System
Steps the system does by itself	—	2
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Payments typed by hand; "paid" without proof
Automation type	Manual	Integration (PayMongo), event-driven (receipt)

M24 Lab Cases

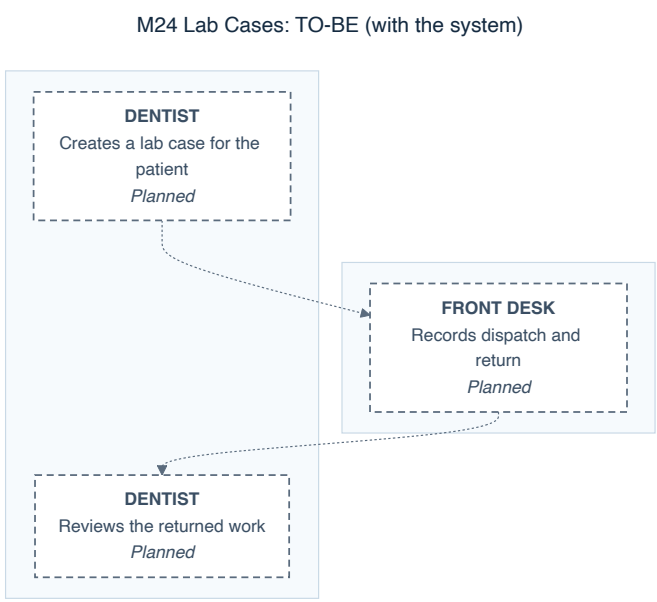
Planned

AS-IS sources: MC 2026-10-12 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

AS-IS (before)



TO-BE (with the system)



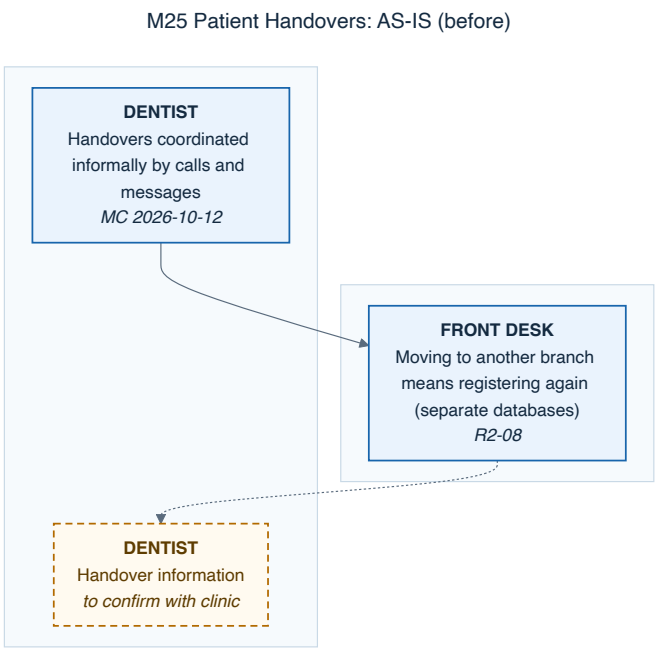
	Before	Now
Steps	2 known (1 to confirm)	0 in the system + 3 planned or mock
Who does it	Front desk, Dentist	Nobody yet (planned)
Steps the system does by itself	—	0
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Not built yet
Automation type	Manual	Planned

M25 Patient Handovers

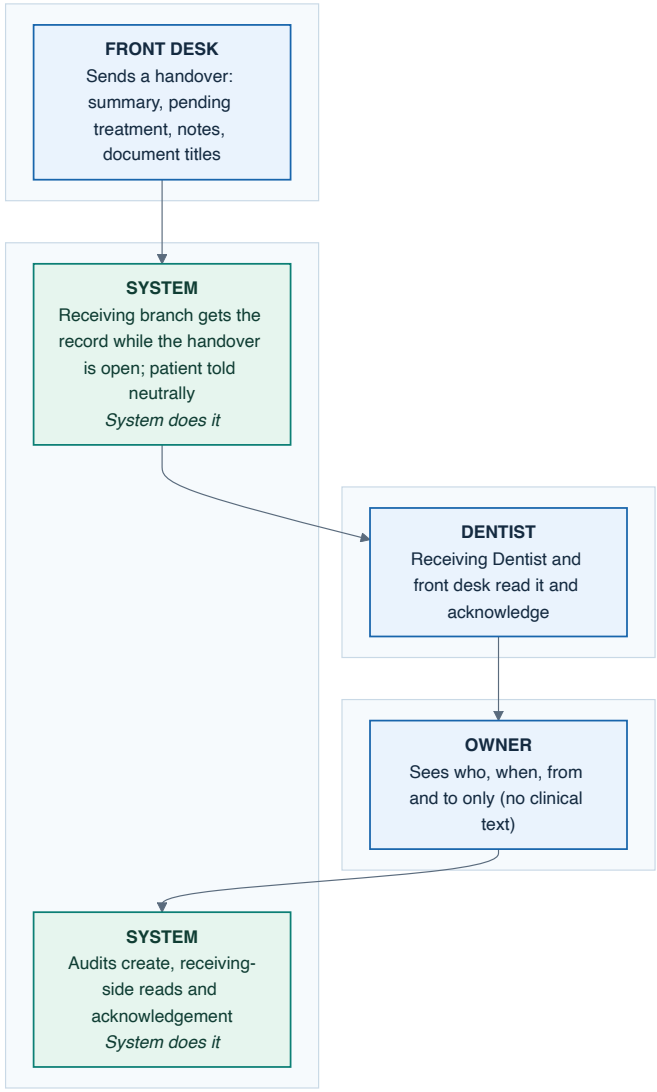
Implemented

AS-IS sources: MC 2026-10-12, R2-08 · dashed amber boxes are to confirm with the clinic (SECOND_INTERVIEW_QUESTIONS.md)

AS-IS (before)



TO-BE (with the system)



	Before	Now
Steps	3 known (1 to confirm)	5 in the system
Who does it	Front desk, Dentist	Front desk, Dentist, Owner, System
Steps the system does by itself	—	2
Estimated time	Not measured (no figure in the interviews)	To be measured in pilot
Errors removed	—	Care details lost when a patient moves branch
Automation type	Manual	Rule-based

Limitations and open items

- Planned modules (M3, M10, M12, M15, M24, M25) and mock-only modules (M19, M20) are drawn as designs, not working software.
- Six scheduled automations (reminders, possible no-shows, overdue invoices, recall, waitlist, Owner summary) run with defaults the clinic has not confirmed; recall ships off with no rules. A no-show is only flagged: a person confirms it.
- Online times are limited by the dentists' seeded shifts until the clinic gives real schedules (PI-06).
- Branches 1 and 2 share one phone number (to confirm); branch 3 has no email and an incomplete address.
- Payments run in PayMongo test mode; live payments need the clinic's own PayMongo account.
- Privacy Policy and Terms are drafts; retention periods are not decided (RA 10173).
- Some timestamps are stored 8 hours early (docs/TIMEZONE_PLAN.md); fix planned after the defense.
- 28 questions for the second interview: docs/defense/SECOND_INTERVIEW_QUESTIONS.md. Full list of clinic decisions: docs/defense/OPEN_ITEMS.md.